Deep Learning for Computer Vision and Mobility Applications

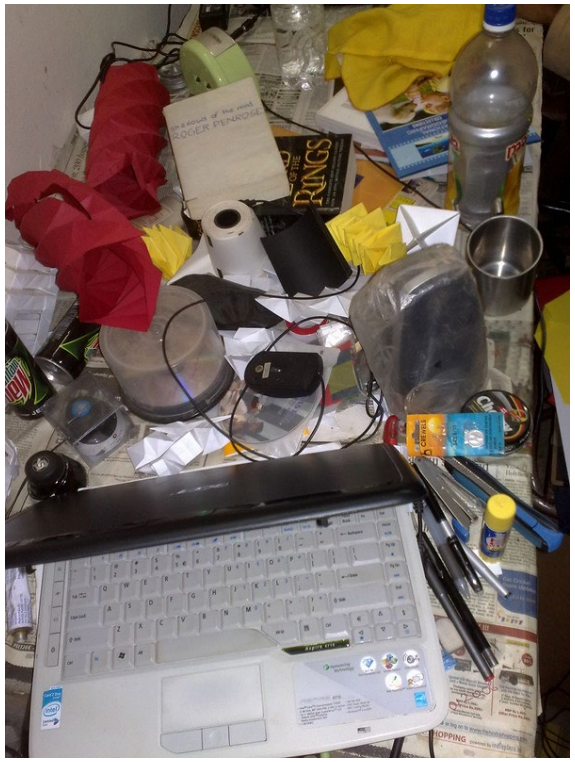
Chaehwa Yoo
School of Electrical Engineering
Chungbuk National University, Korea
E-mail: chaehway@chungbuk.ac.kr



Today's Topic

- What is Computer Vision?
- A brief history of Computer Vision and Deep Learning
- Computer Vision Applications
  - Perceiving and Understanding the Visual World
  - Generative and Interactive Visual Intelligence
  - Human-Centered Applications and Implications
- Remaining Challenges

What is Computer Vision?



Where is the gluestick?
Find the book – what's its full title?

Credit: Bharath Kishore, Flickr CC License

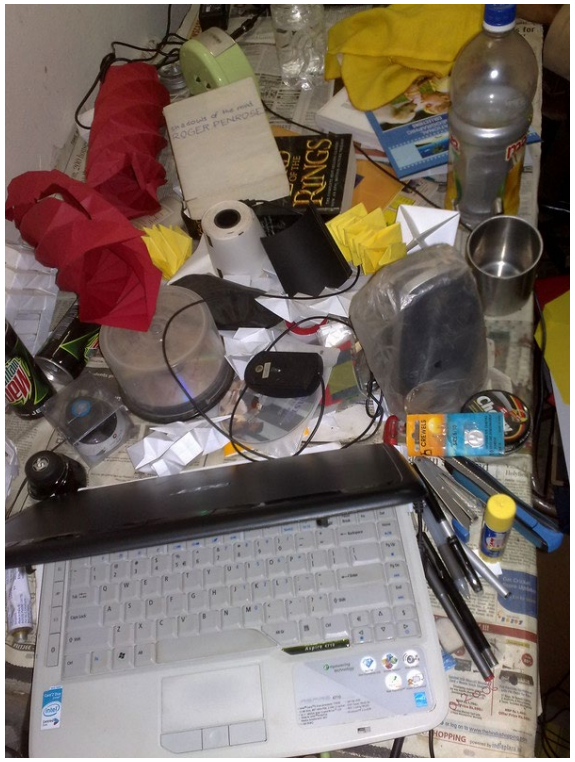


What is wrong with this image?

Credit: Erik Johansson

Slide inspiration: Vineeth N Balasubramanian

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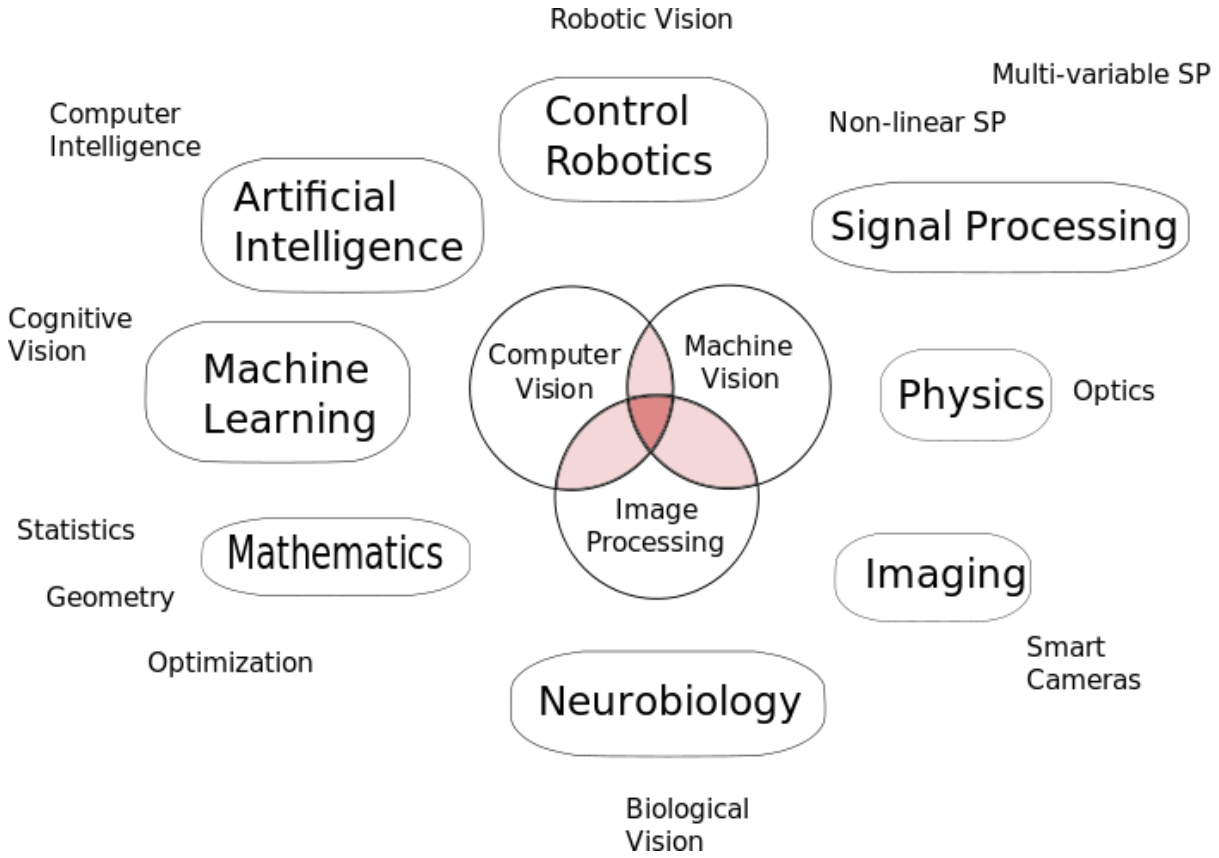
Can a machine answer the above questions?

Slide inspiration: Vineeth N Balasubramanian

What is Computer Vision?

Computer Vision (CV)

- A field that seeks to automate and endow a computing framework with the ability to interpret images the way humans do.
- A sub-topic of Artificial Intelligence



Slide inspiration: Vineeth N Balasubramanian

Computer Vision is everywhere!







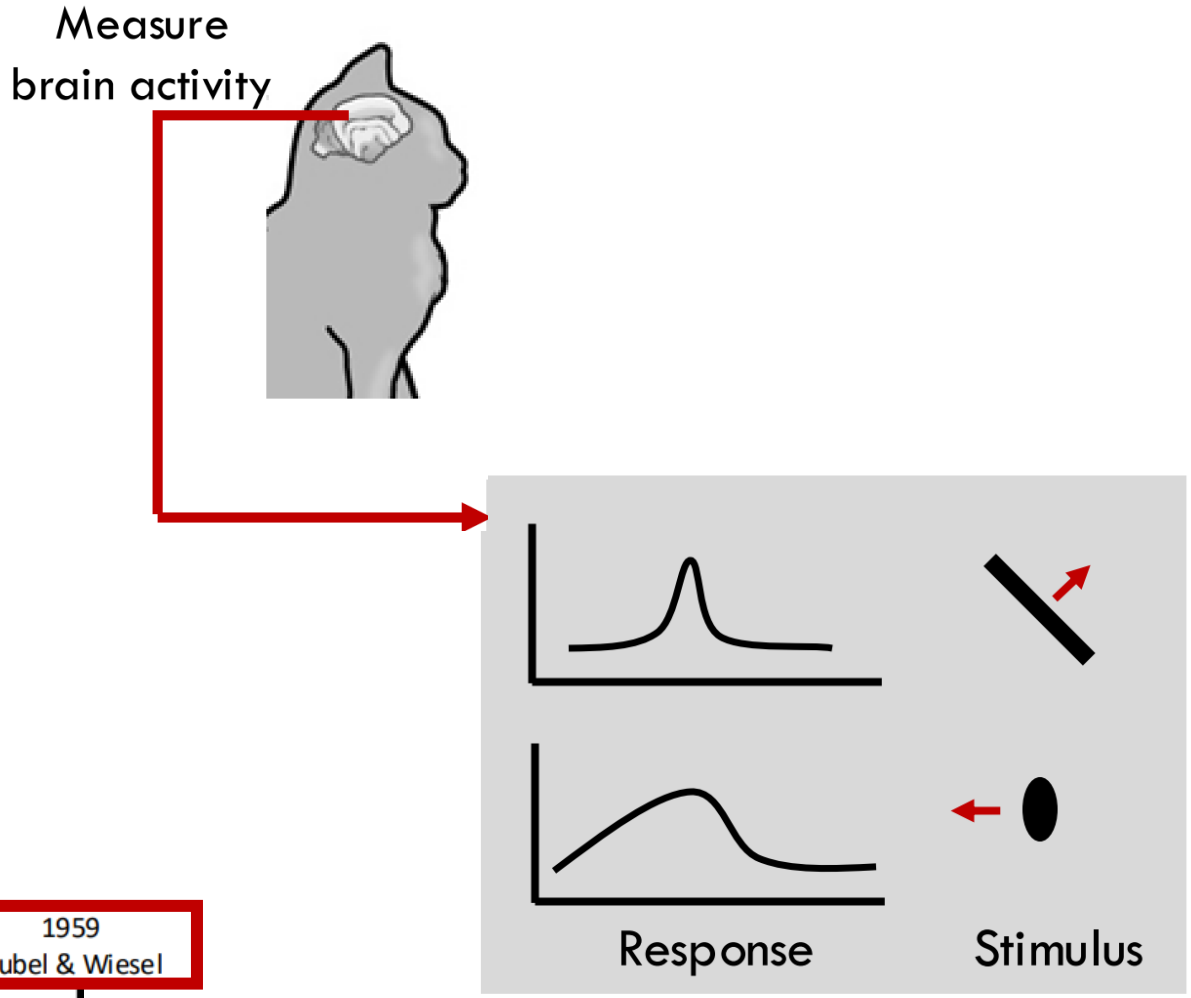


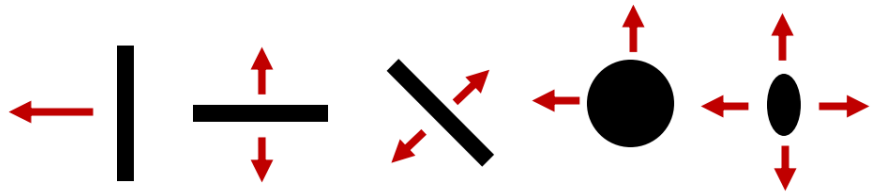





Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Hubel and Wiesel, 1959





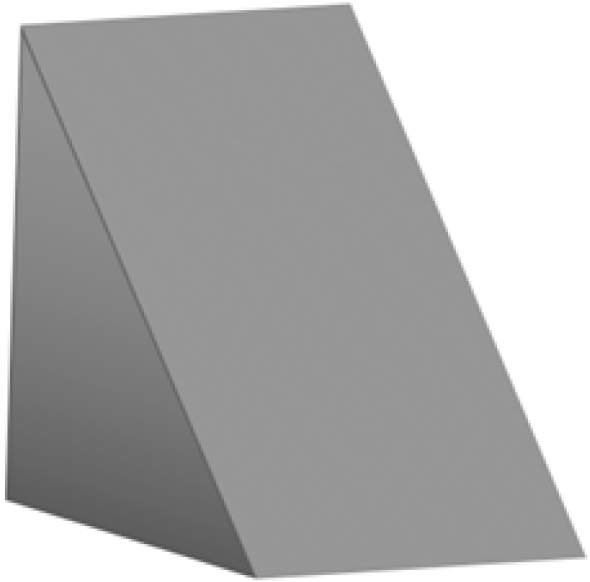
Simple cells:
Response to specific rotation and orientation

Complex cells:
Response to light orientation and movement, some translation invariance

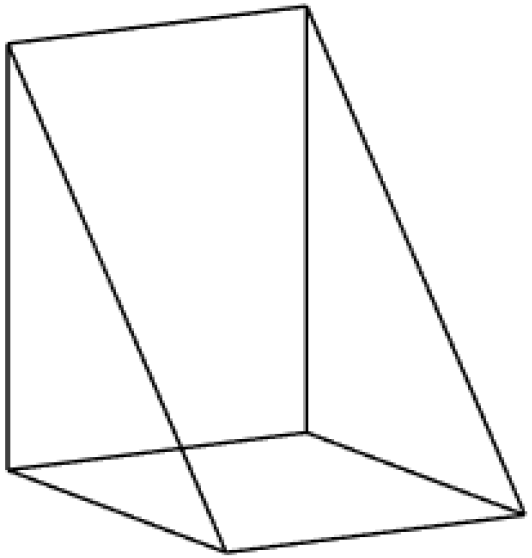


Slide inspiration: Justin Johnson

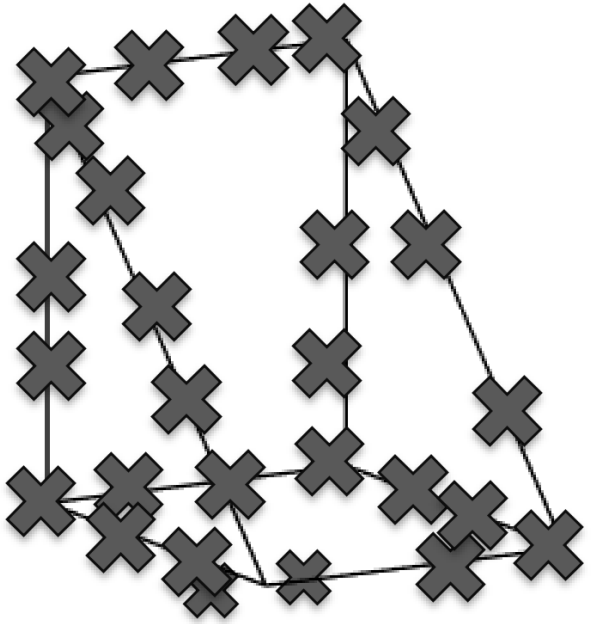
Larry Roberts, 1963



(a) Original picture

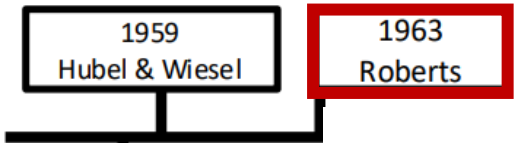


(b) Differentiated picture



(c) Feature points selected

Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963



Slide inspiration: Justin Johnson

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

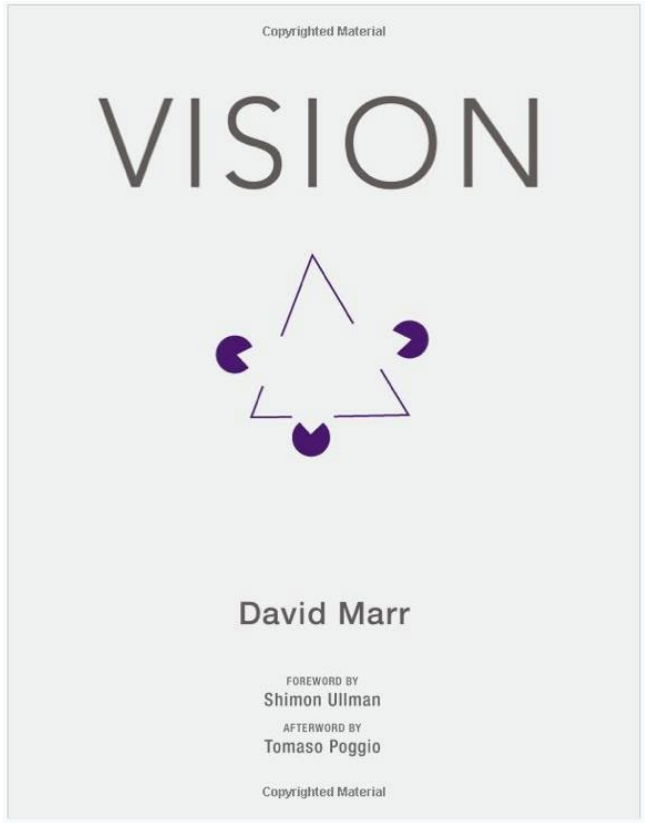
Seymour Papert

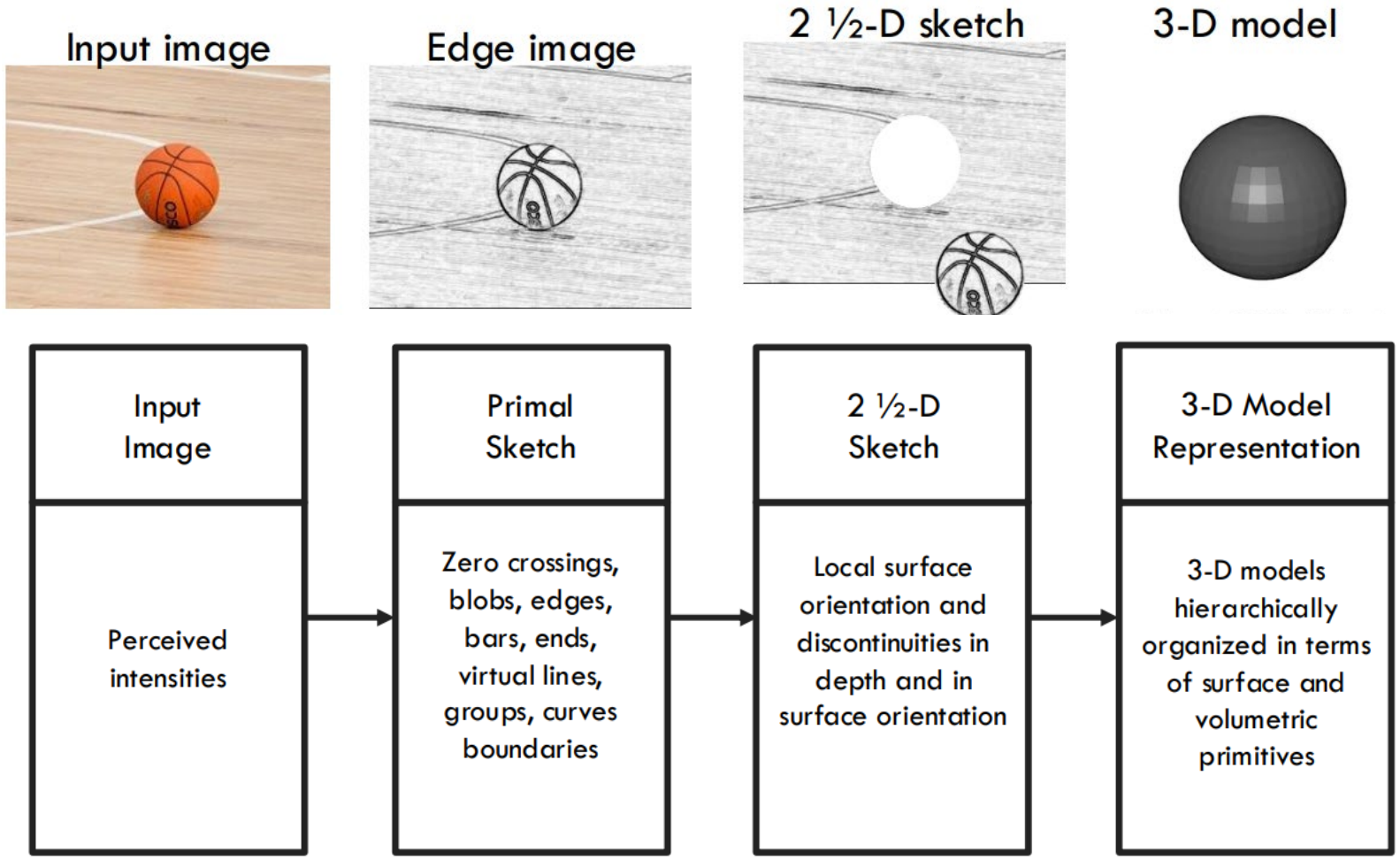
The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

1959
Hubel & Wiesel

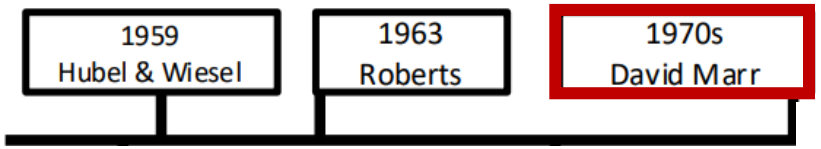
1963
Roberts

Slide inspiration: Justin Johnson





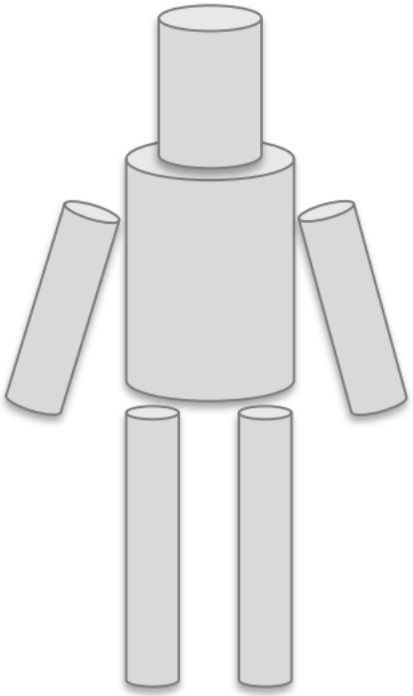
Stages of Visual Representation, David Marr, 1970s

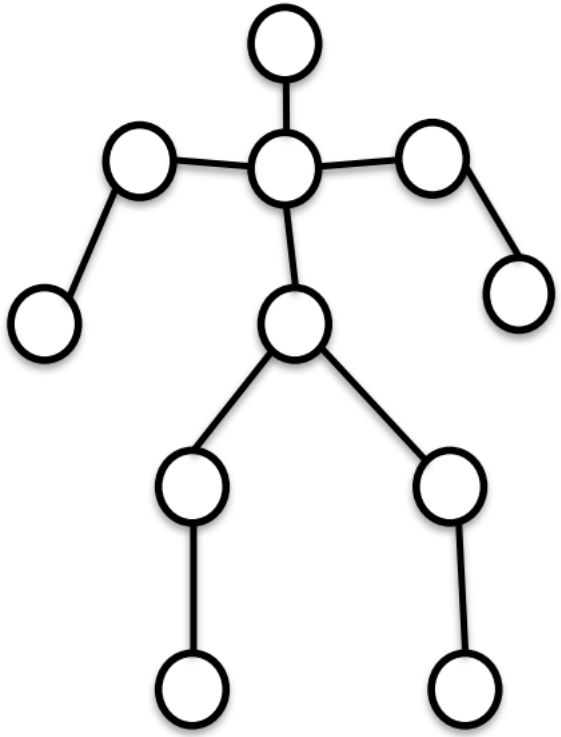


Slide inspiration: Justin Johnson

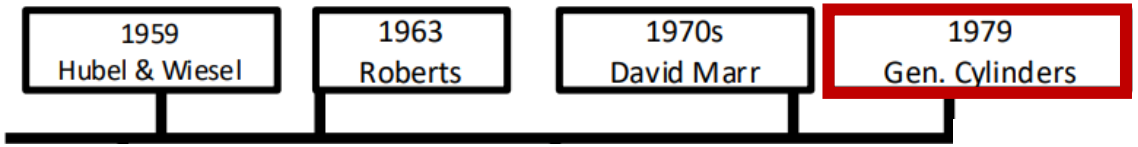
Recognition via Parts (1970s)

Generalized Cylinders, Brooks and Binford, 1979





Pictorial Structures, Fischler and Elshlager, 1973



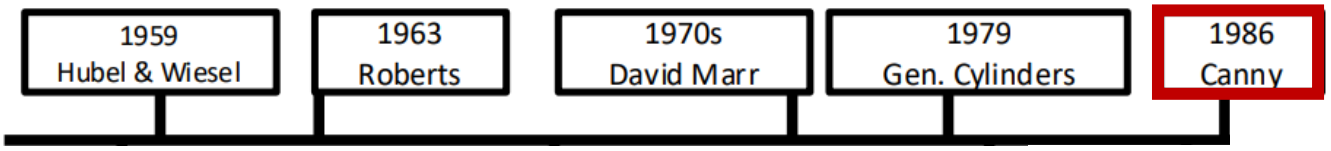
Slide inspiration: Justin Johnson

Recognition via Edge Detection (1980s)





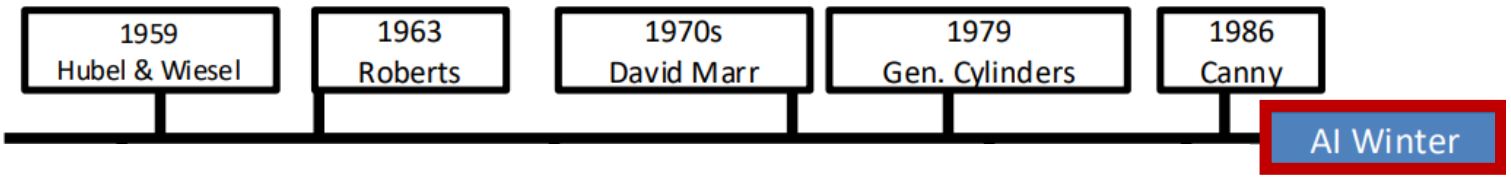
John Canny, 1986
David Lowe, 1987



Slide inspiration: Justin Johnson

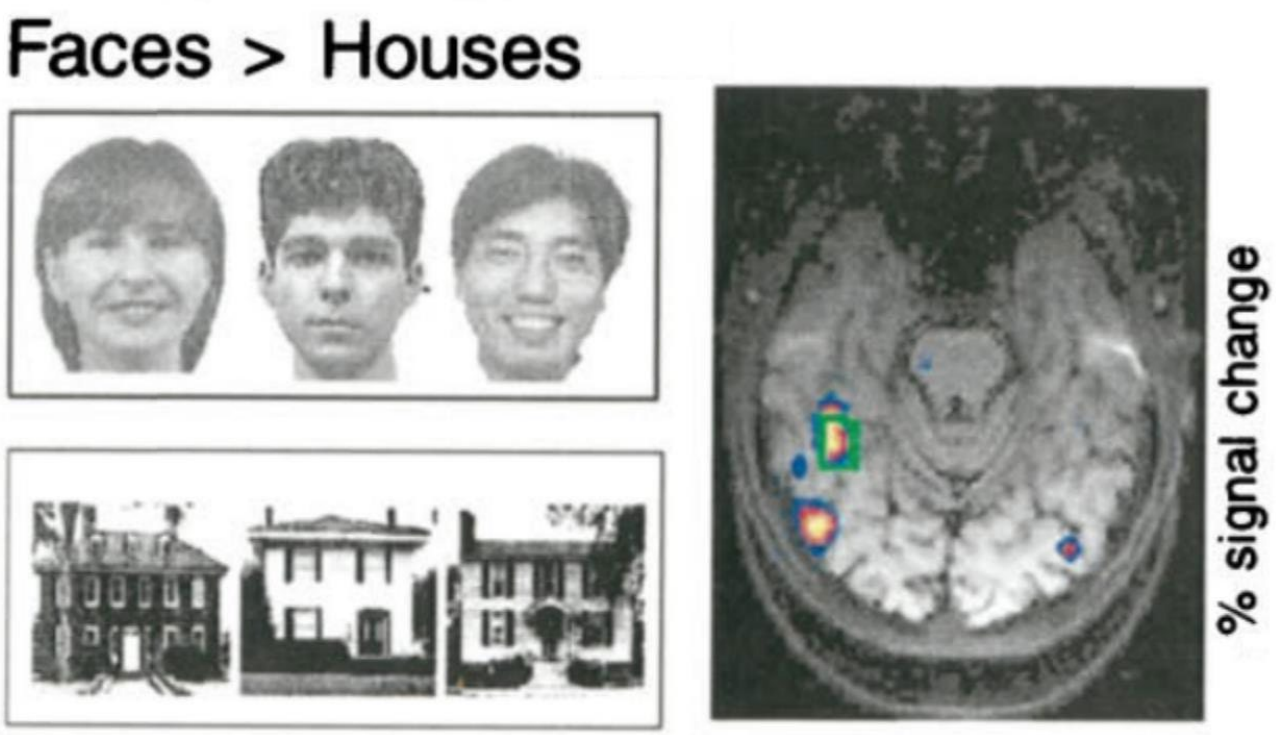
Arriving at an "AI winter"

- Enthusiasm for AI research dwindled
- "Expert Systems" failed to deliver on their promises
- But subfields of AI continues to grow
- Computer vision, NLP, robotics, compbio, etc.

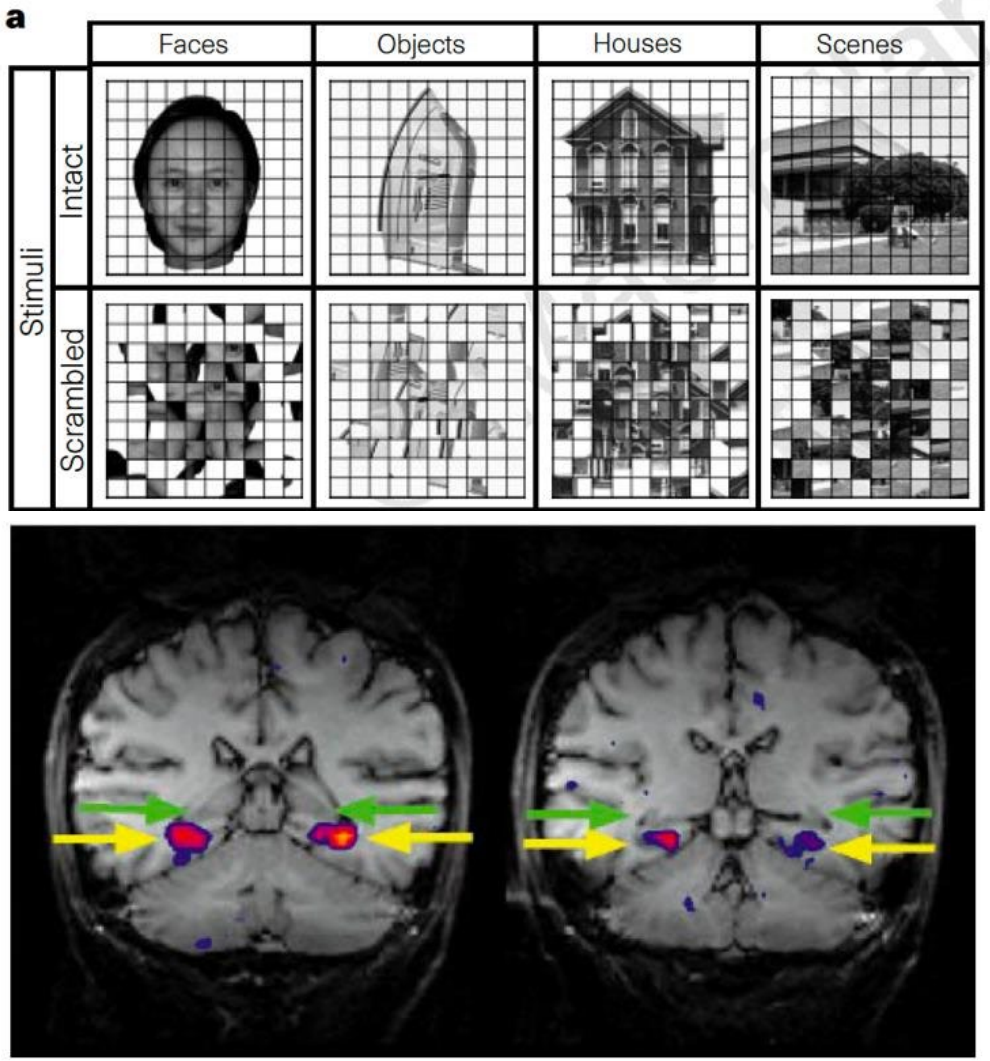


Slide inspiration: Justin Johnson

Neural correlates of object & scene recognition



Kanwisher et al. J. Neuro. 1997

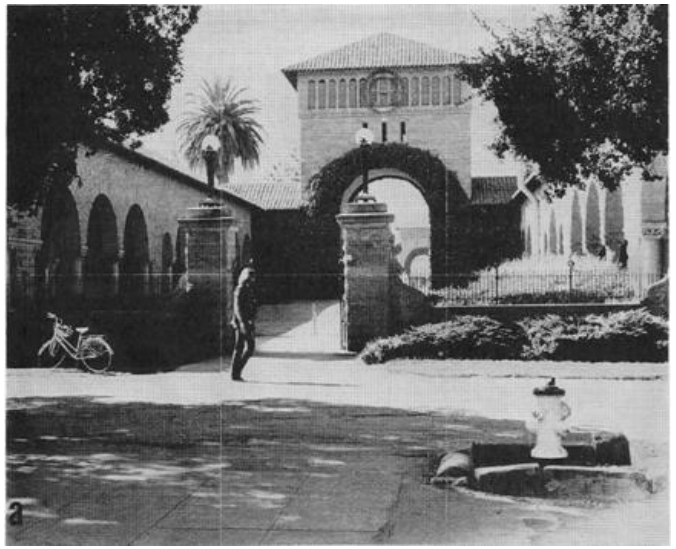


Epstein & Kanwisher, Nature, 1998

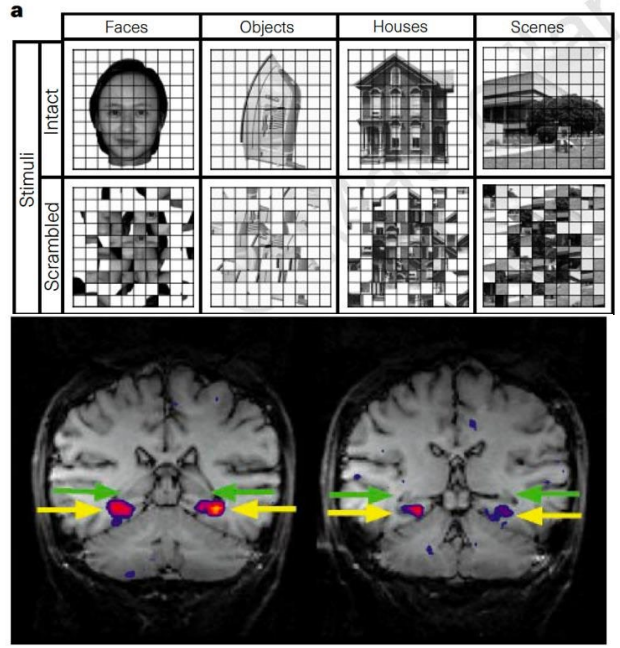
Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

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Visual recognition is a fundamental task for visual intelligence





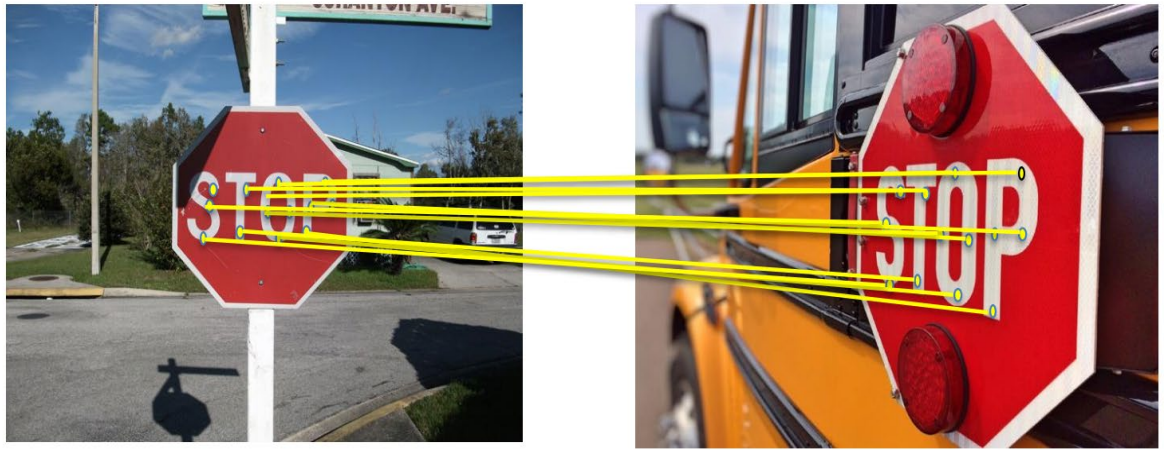


Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

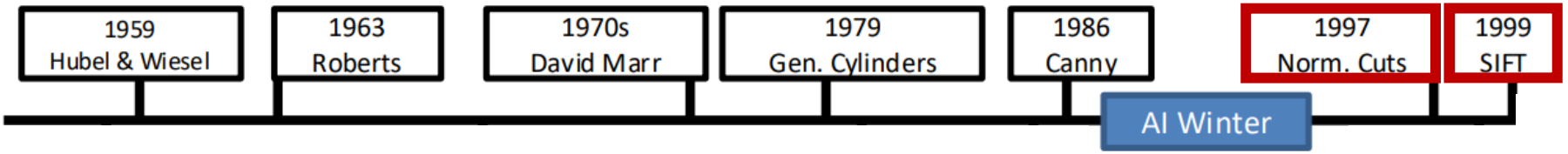
Recognition via Grouping (1990s) and Matching (2000s)



Normalized Cuts, Shi and Malik, 1997



SIFT, David Lowe, 1999



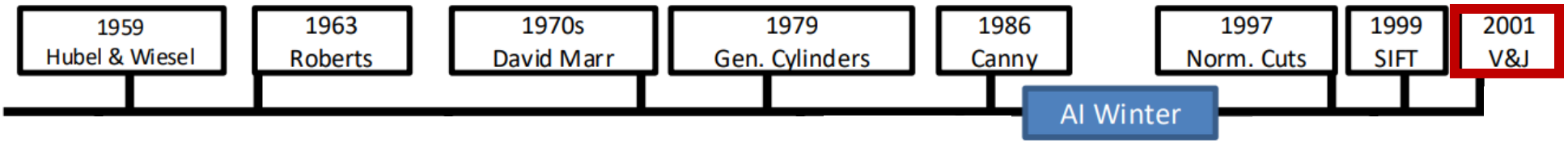
Slide inspiration: Justin Johnson

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision





Slide inspiration: Justin Johnson

Caltech 101 images



PASCAL Visual Object Challenge

Image is CC0 1.0 public domain



Image is CC0 1.0 public domain

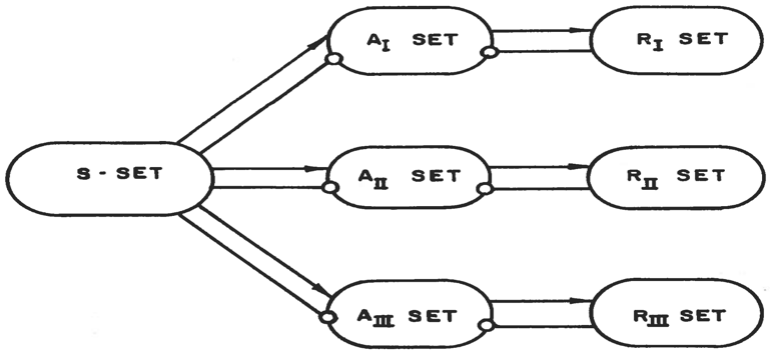
1959 Hubel & Wiesel
1963 Roberts
1970s David Marr
1979 Gen. Cylinders
1986 Canny
AI Winter
1997 Norm. Cuts
1999 SIFT
2001 V&J
2004, 2007 Caltech101; PASCAL

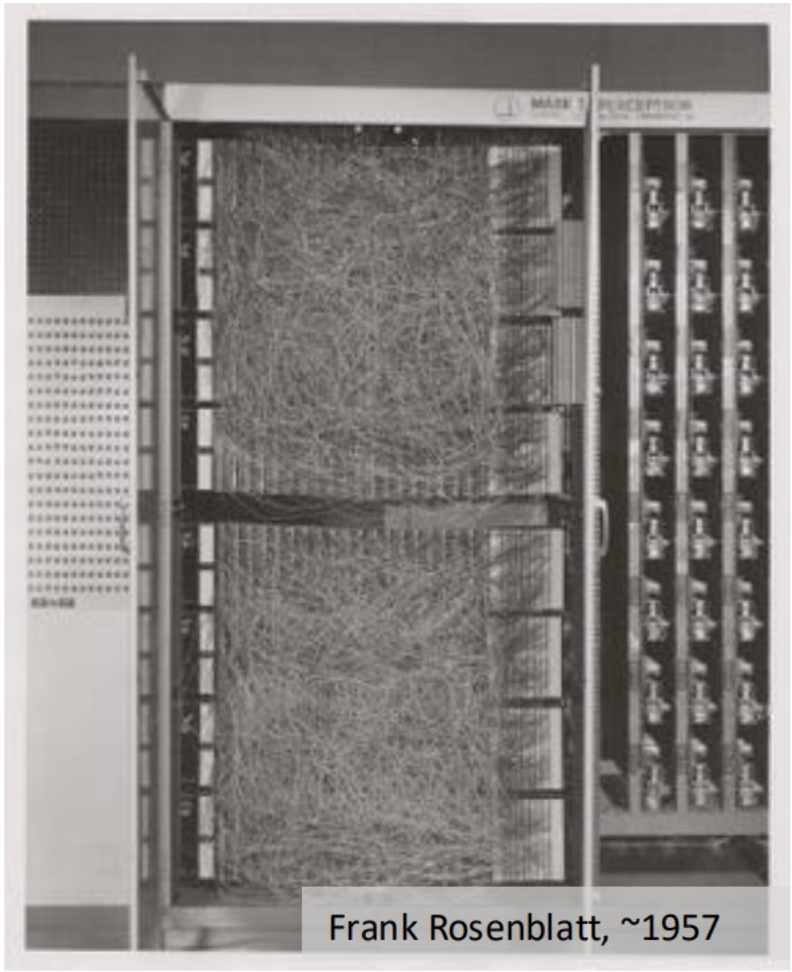
Slide inspiration: Justin Johnson

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton†
& Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California, San Diego, La Jolla, California 92093, USA
† Department of Computer Science, Carnegie-Mellon University, Pittsburgh, Philadelphia 15213, USA



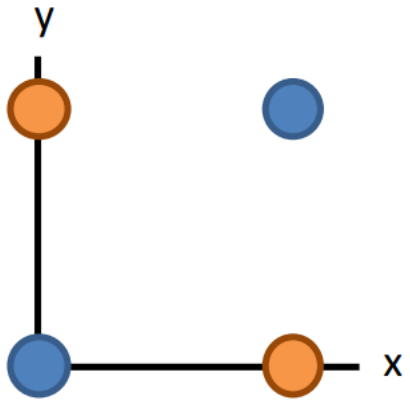

Frank Rosenblatt, ~1957

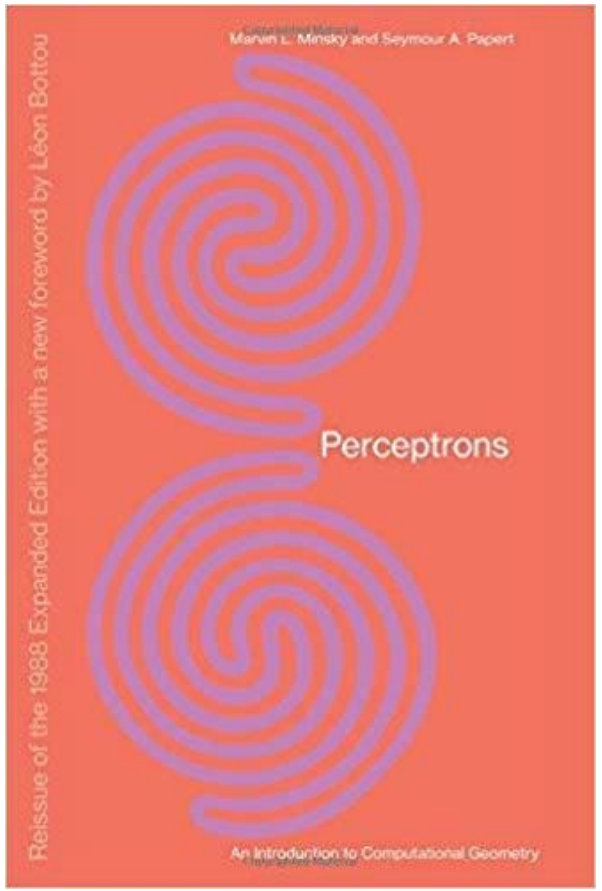
1958 Perceptron
1959 Hubel & Wiesel
1963 Roberts
1970s David Marr
1979 Gen. Cylinders
1986 Canny
AI Winter
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2001 V&J
2004, 2007 Caltech101; PASCAL

Slide inspiration: Justin Johnson

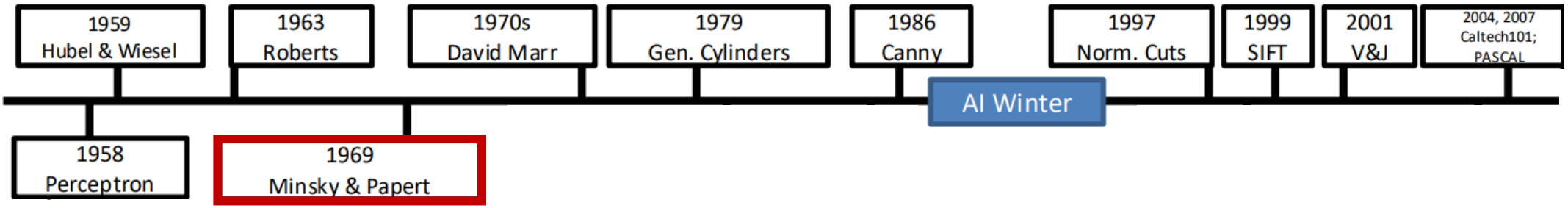
Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0





Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field



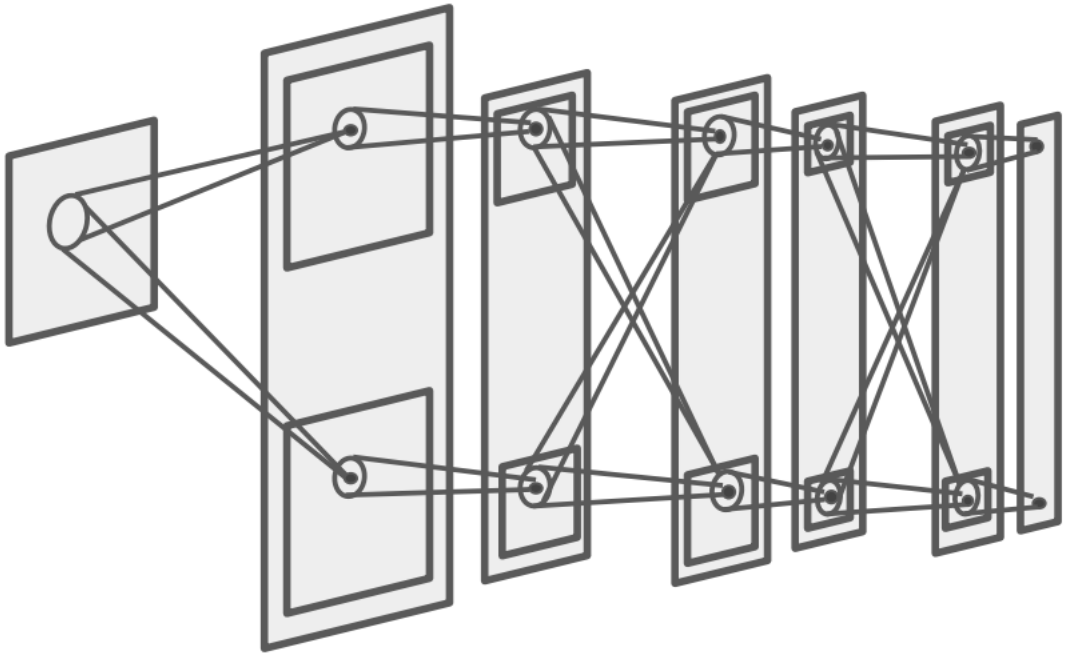
Slide inspiration: Justin Johnson

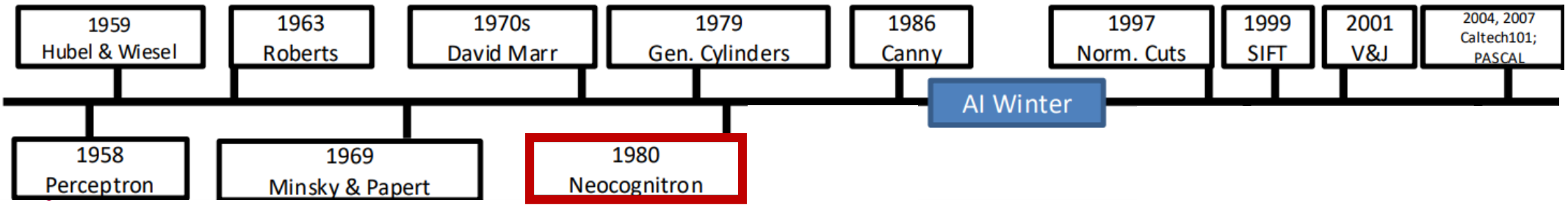
Neocognitron: Fukushima, 1980

Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells

Interleaved simple cells (convolution) and complex cells (pooling)

No practical training algorithm





Slide inspiration: Justin Johnson

Backprop: Rumelhart, Hinton, and Williams, 1986

Introduced backpropagation for computing gradients in neural networks

Successfully trained perceptrons with multiple layers

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$

recognizable math

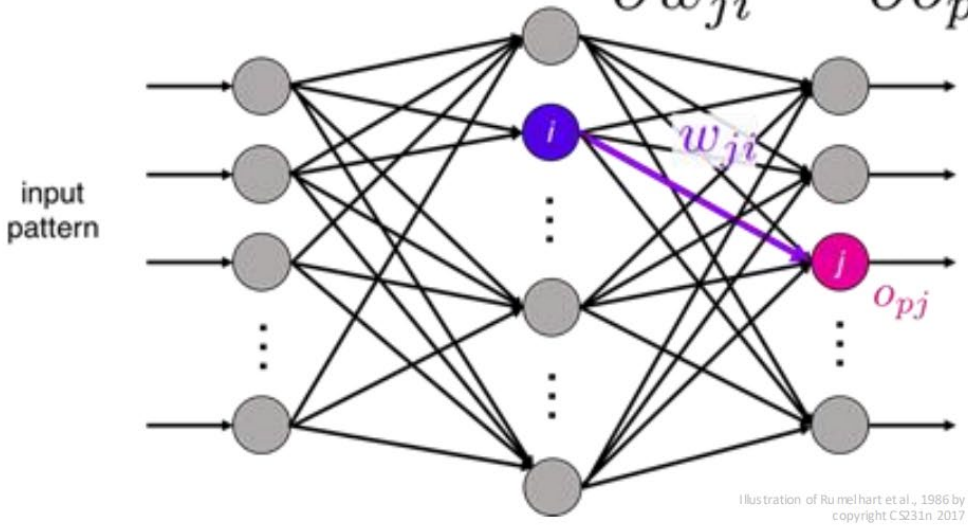
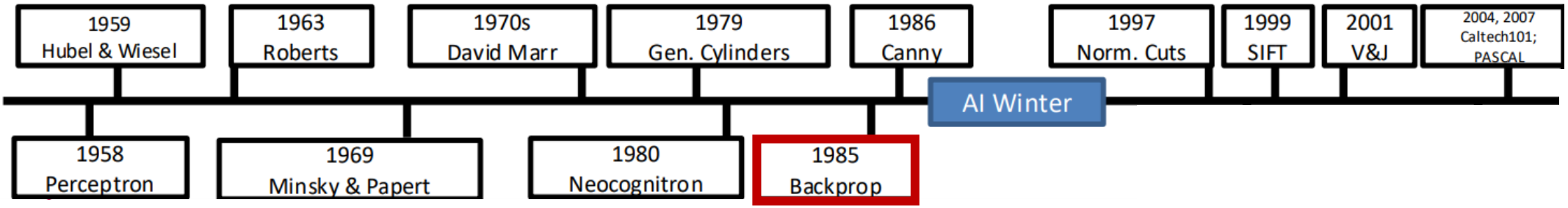
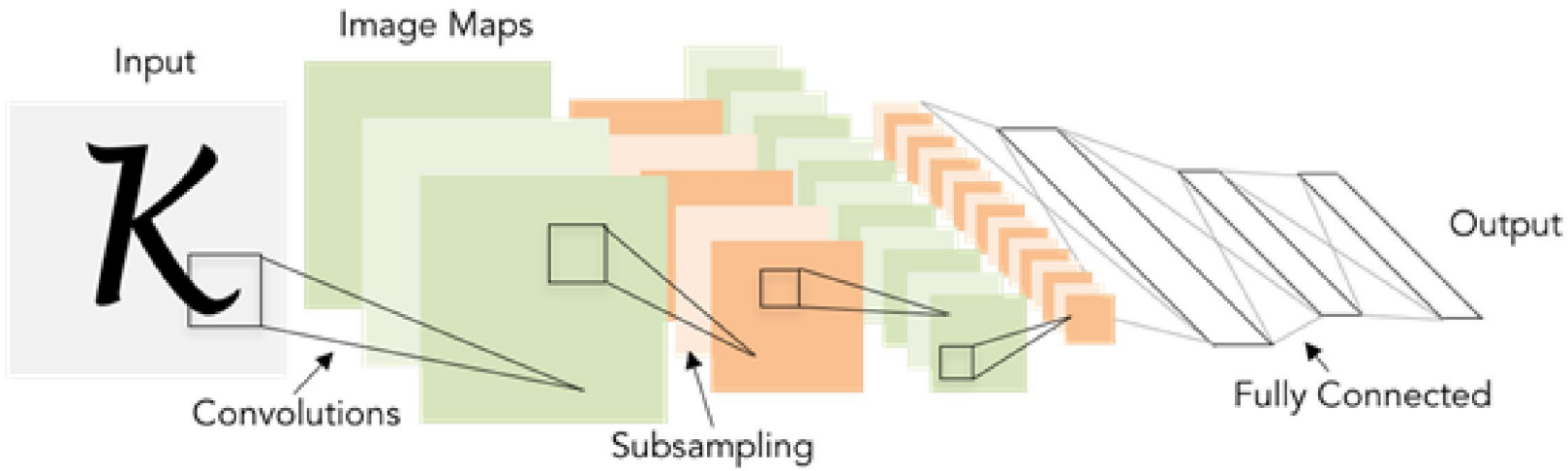


Illustration of Rumelhart et al., 1986 by Lane McIntosh, copyright CS231n 2017



Slide inspiration: Justin Johnson

Convolutional Networks: LeCun et al, 1998

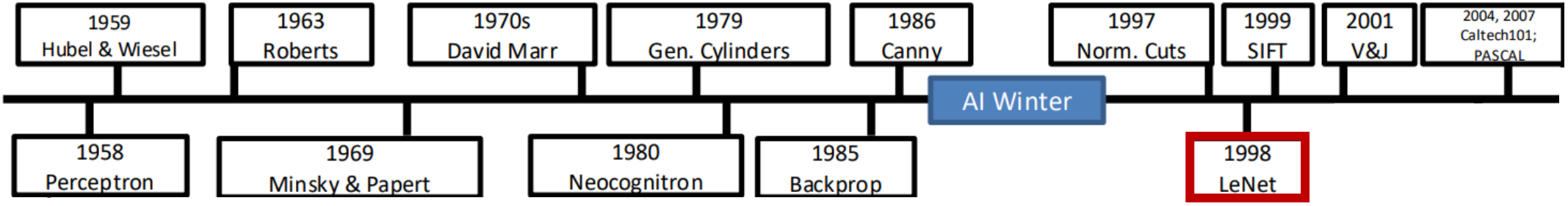


Applied backprop algorithm to a Neocognitron–like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!



Slide inspiration: Justin Johnson

AT&T

LeNet 5

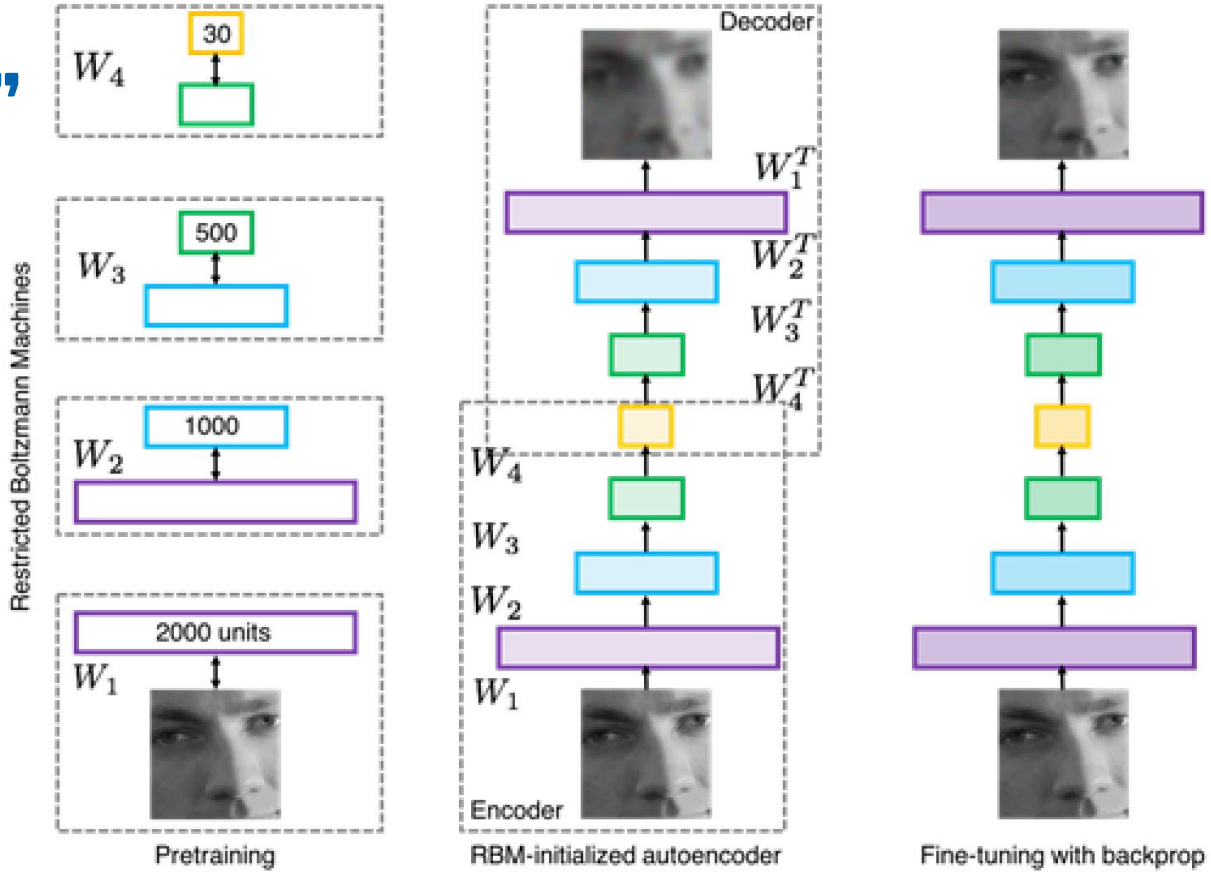
RESEARCH

answer: 0

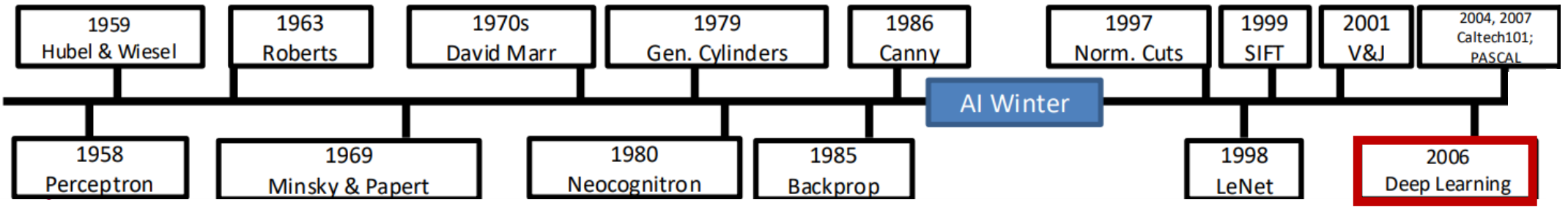
2000s: "Deep Learning"

People tried to train neural networks that were deeper and deeper

Not a mainstream research topic at this time



Hinton and Salakhutdinov, 2006
Bengio et al, 2007
Lee et al, 2009
Glorot and Bengio, 2010



Slide inspiration: Justin Johnson

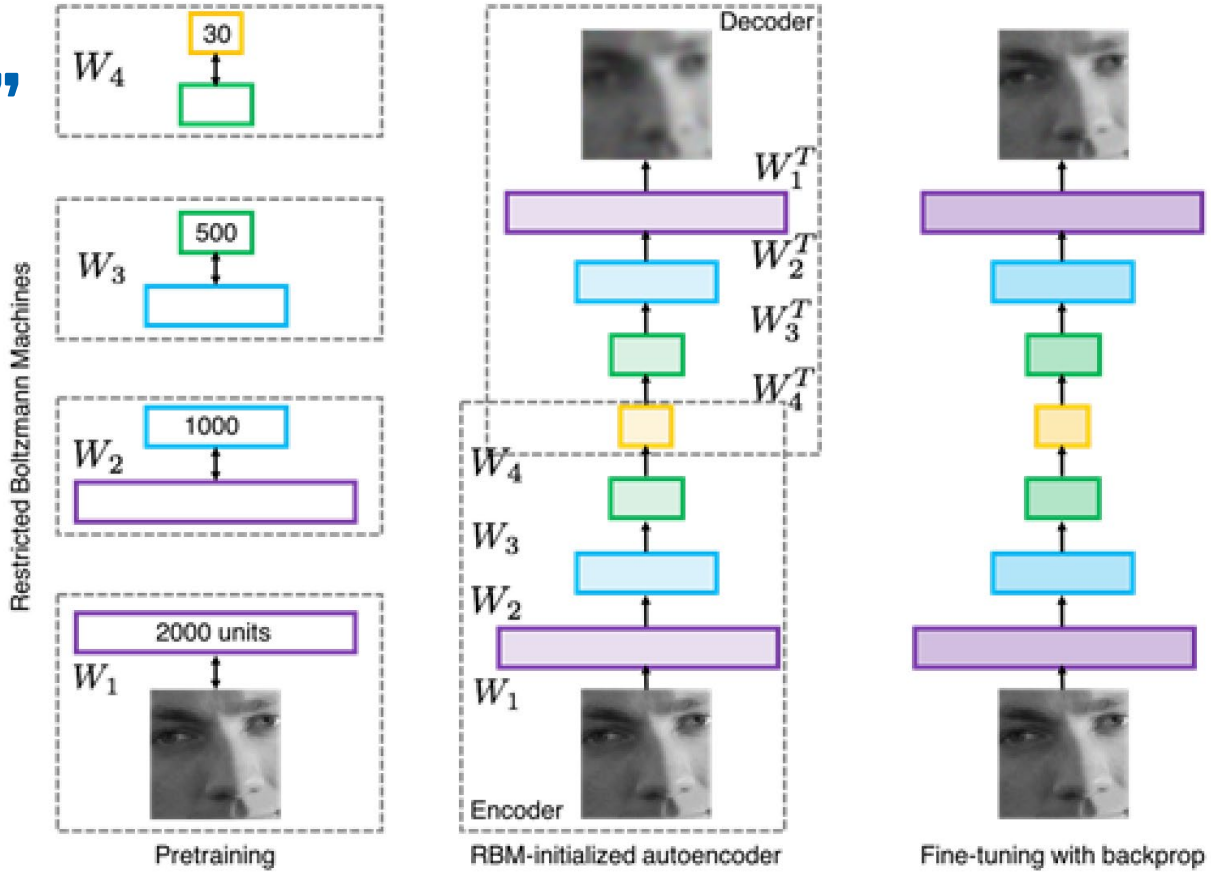
2000s: "Deep Learning"

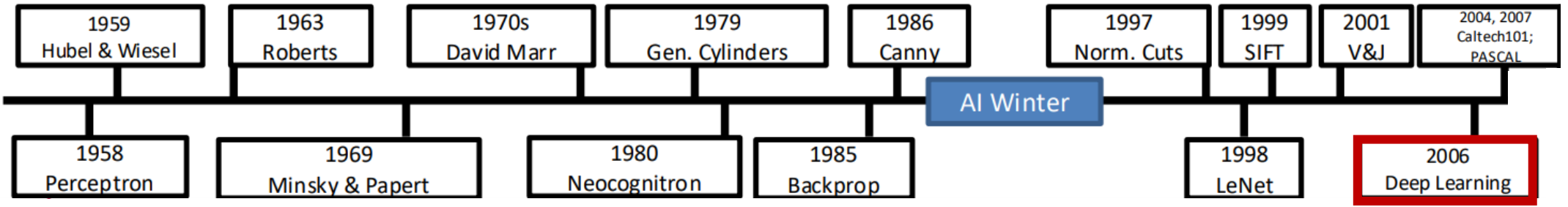
People tried to train neural networks that were deeper and deeper

Not a mainstream research topic at this time

No good dataset to work on

Hinton and Salakhutdinov, 2006
Bengio et al, 2007
Lee et al, 2009
Glorot and Bengio, 2010





Slide inspiration: Justin Johnson

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes

1,431,167 images



Output:

Scale

T-shirt

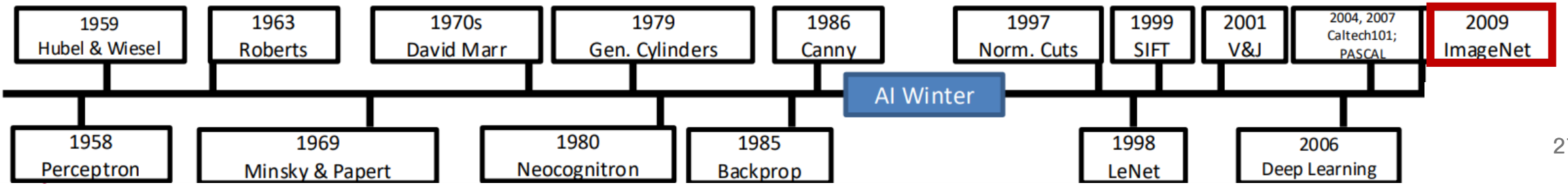
Steel drum

Drumstick

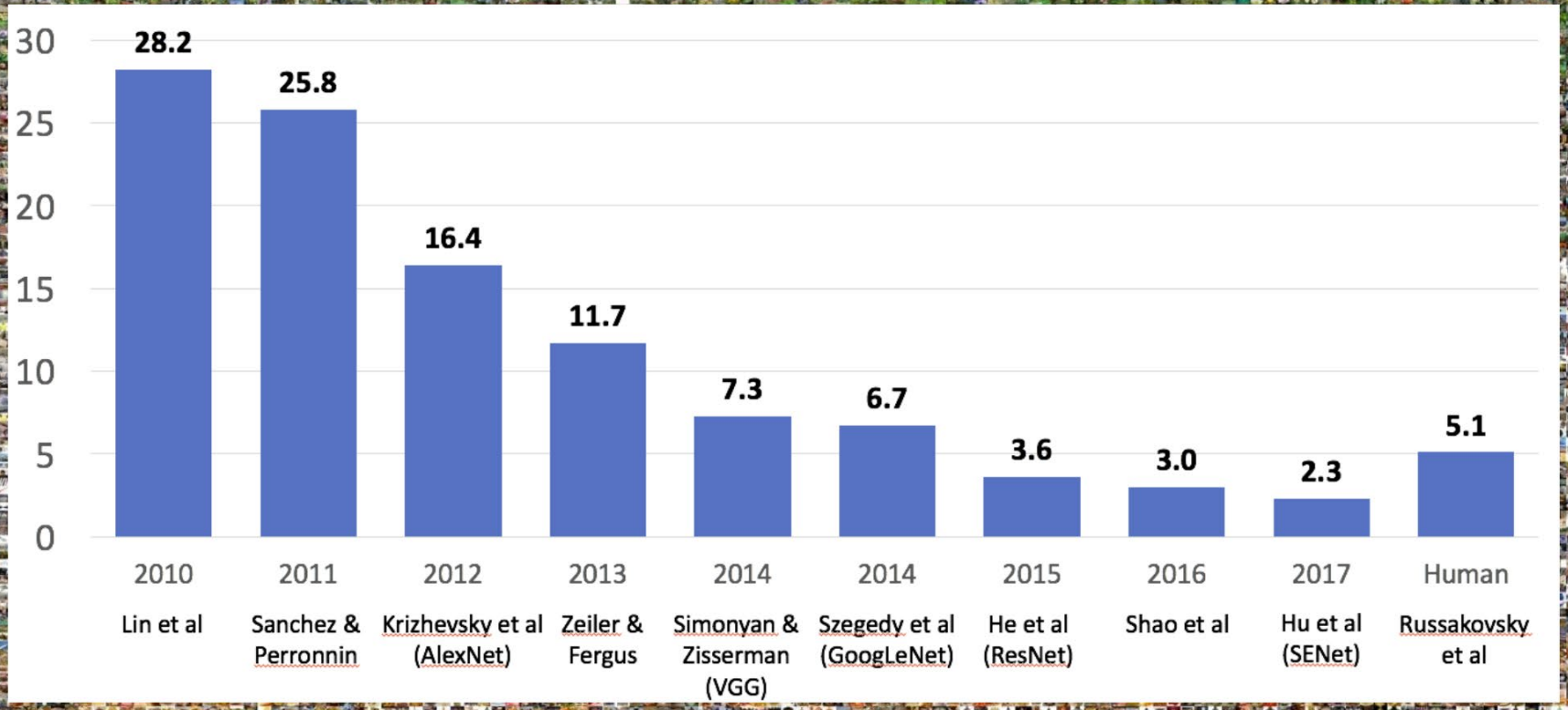
Mud turtle

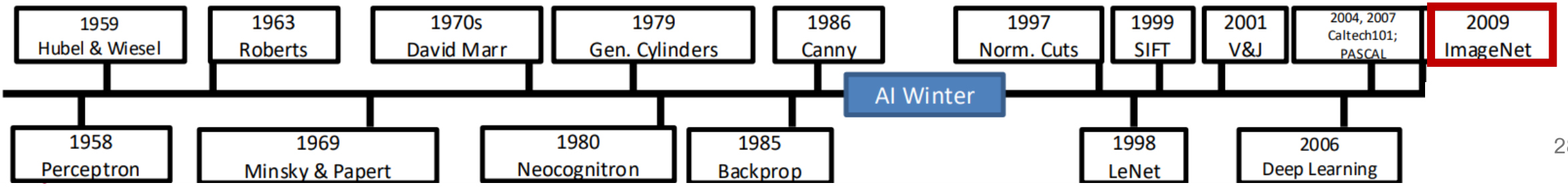
Deng et al, 2009

Russakovsky et al. IJCV 2015

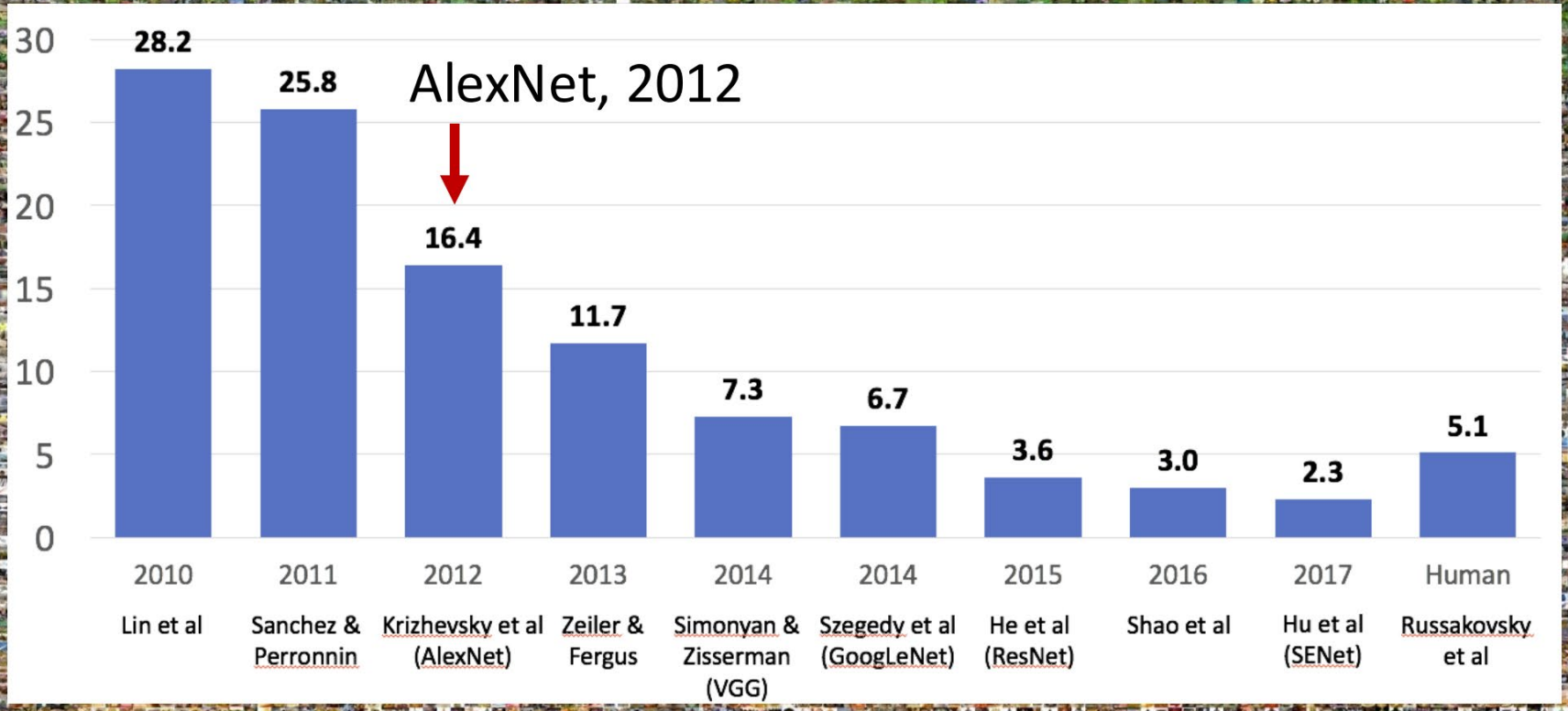


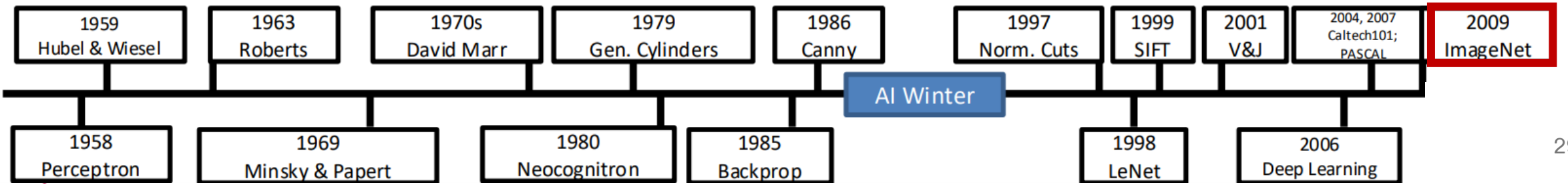
IMAGENET Large Scale Visual Recognition Challenge



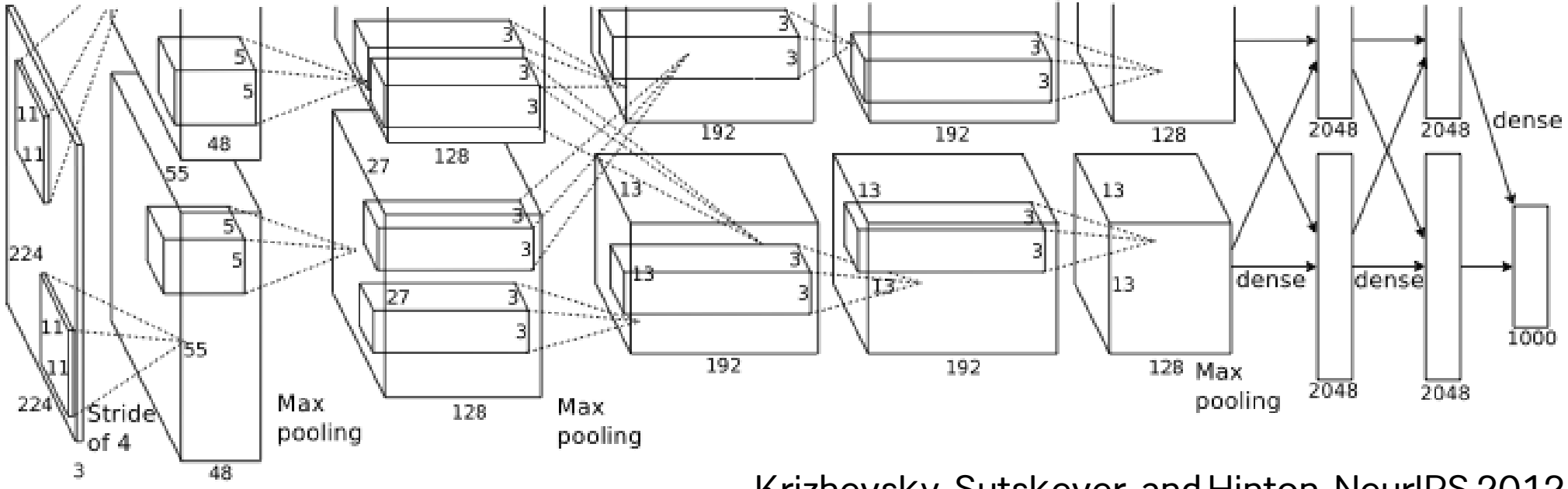


IMAGENET Large Scale Visual Recognition Challenge

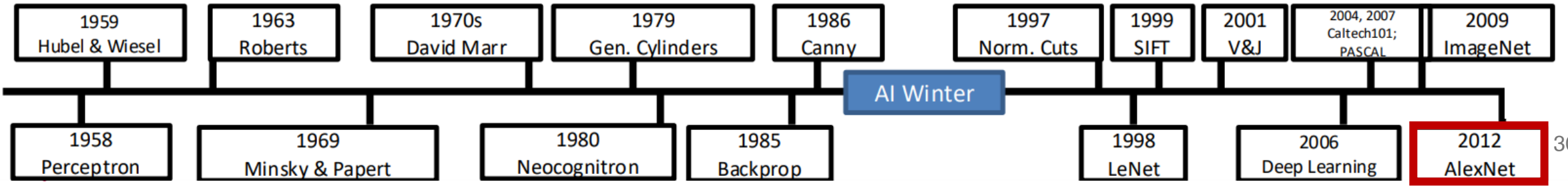




AlexNet: Deep Learning Goes Mainstream

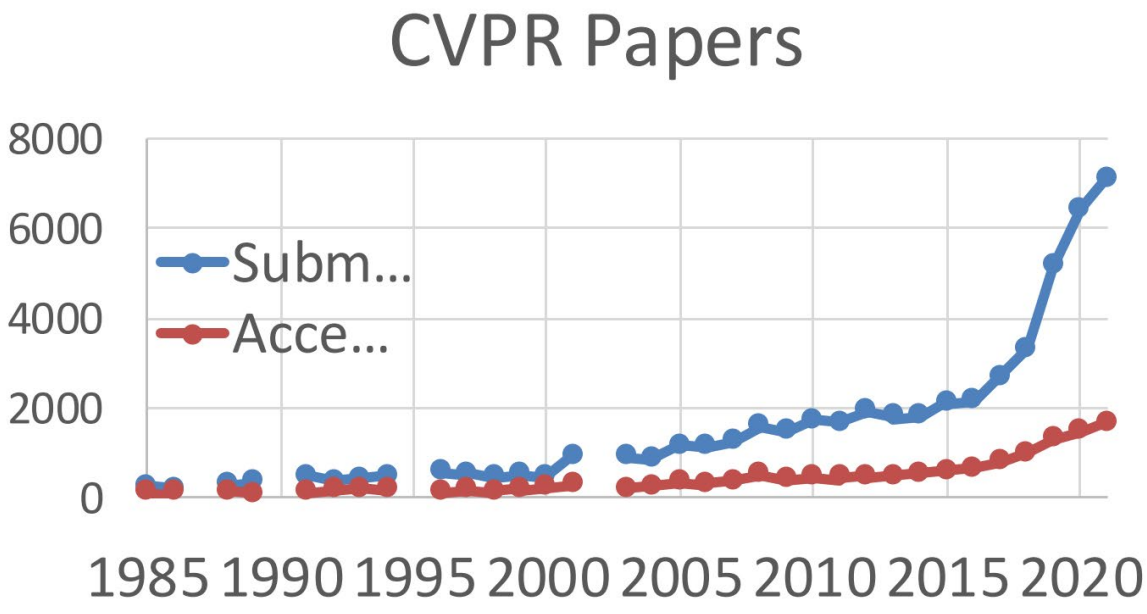


Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

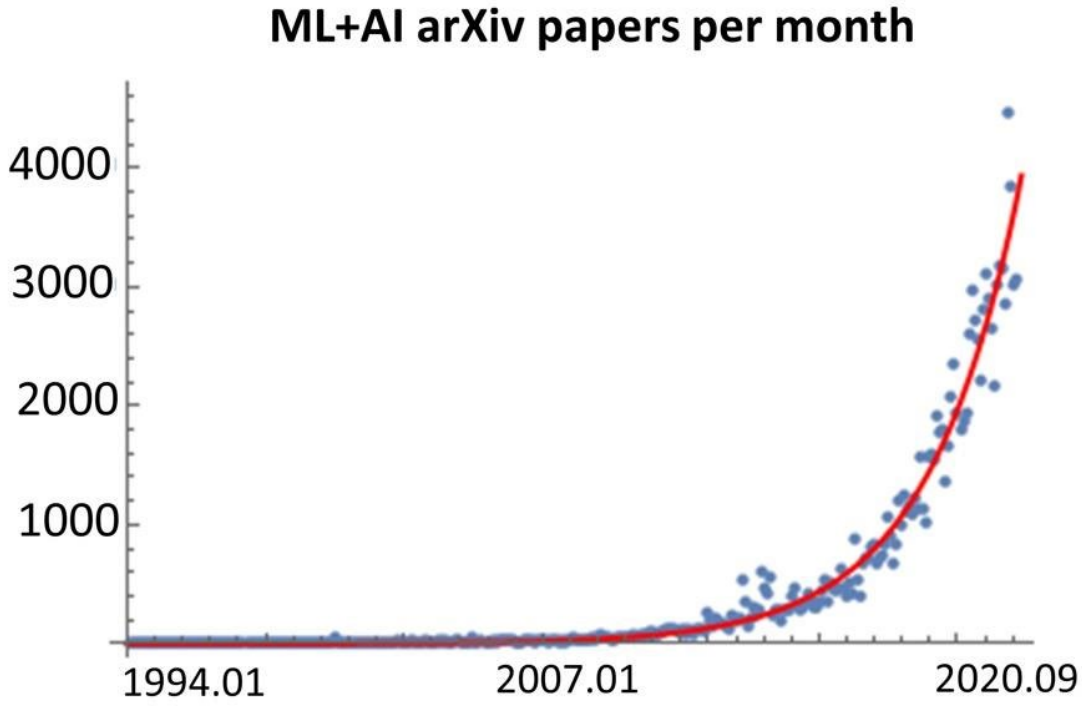


Slide inspiration: Justin Johnson

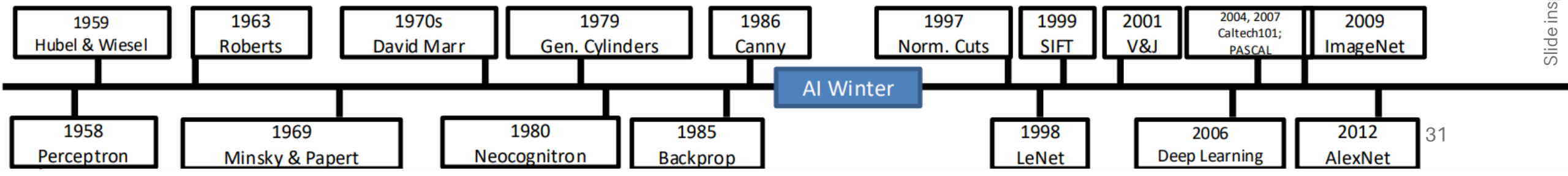
2012 to Present: Deep Learning Explosion



Publications at top Computer Vision conference



arXiv papers per month

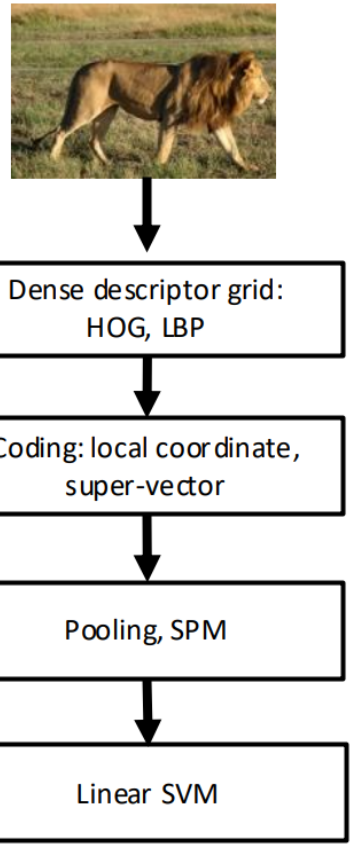


Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Year 2010

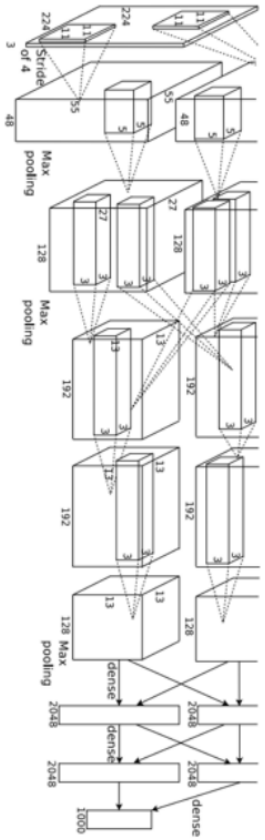
SVM



[Lin CVPR 2011]

Year 2012

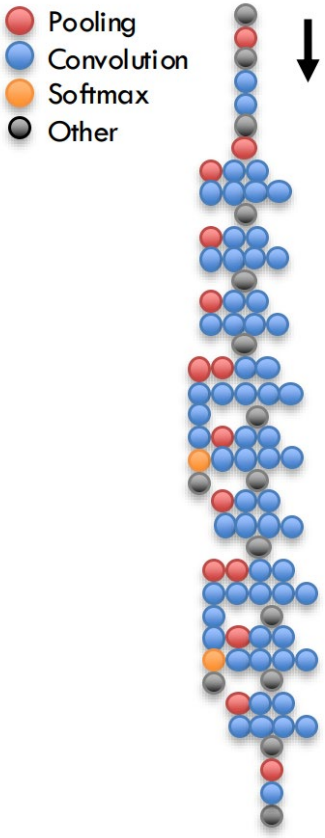
AlexNet



[Krizhevsky NIPS 2012]

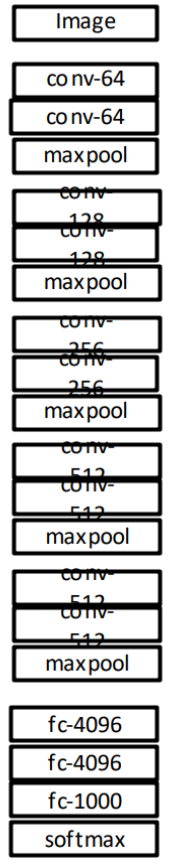
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

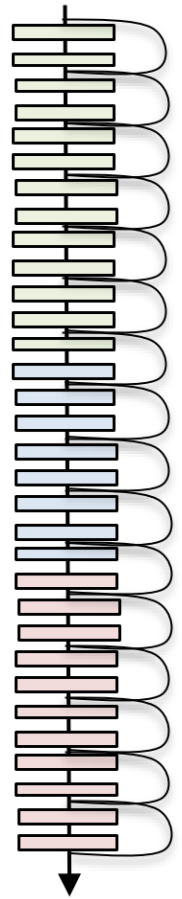
VGGNet



[Simonyan arxiv 2014]

Year 2015

ResNet



[He ICCV 2015]

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

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2012 to Present: Deep Learning is Everywhere

Image Classification



Image Retrieval



Slide inspiration: Justin Johnson

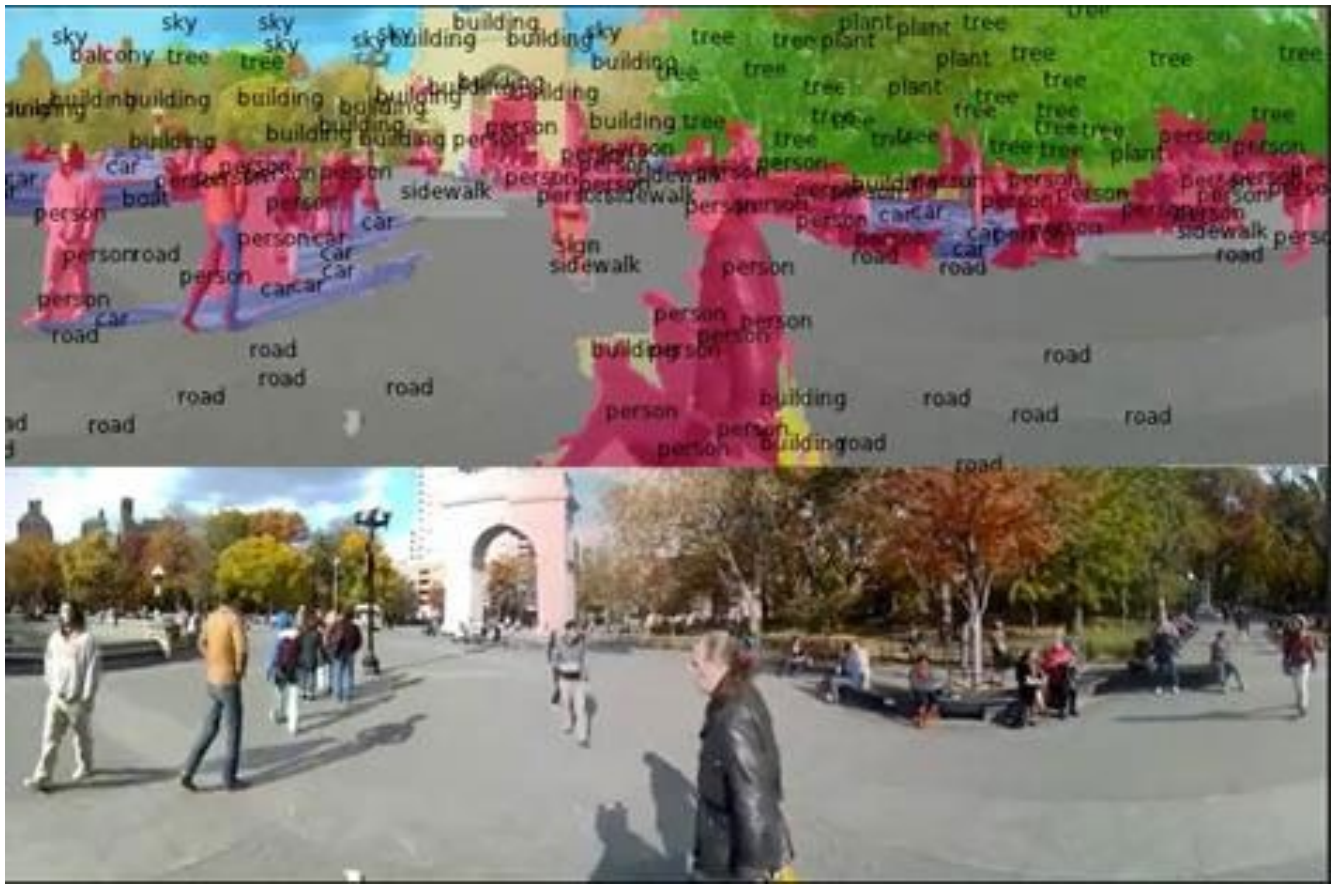
2012 to Present: Deep Learning is Everywhere

Object detection

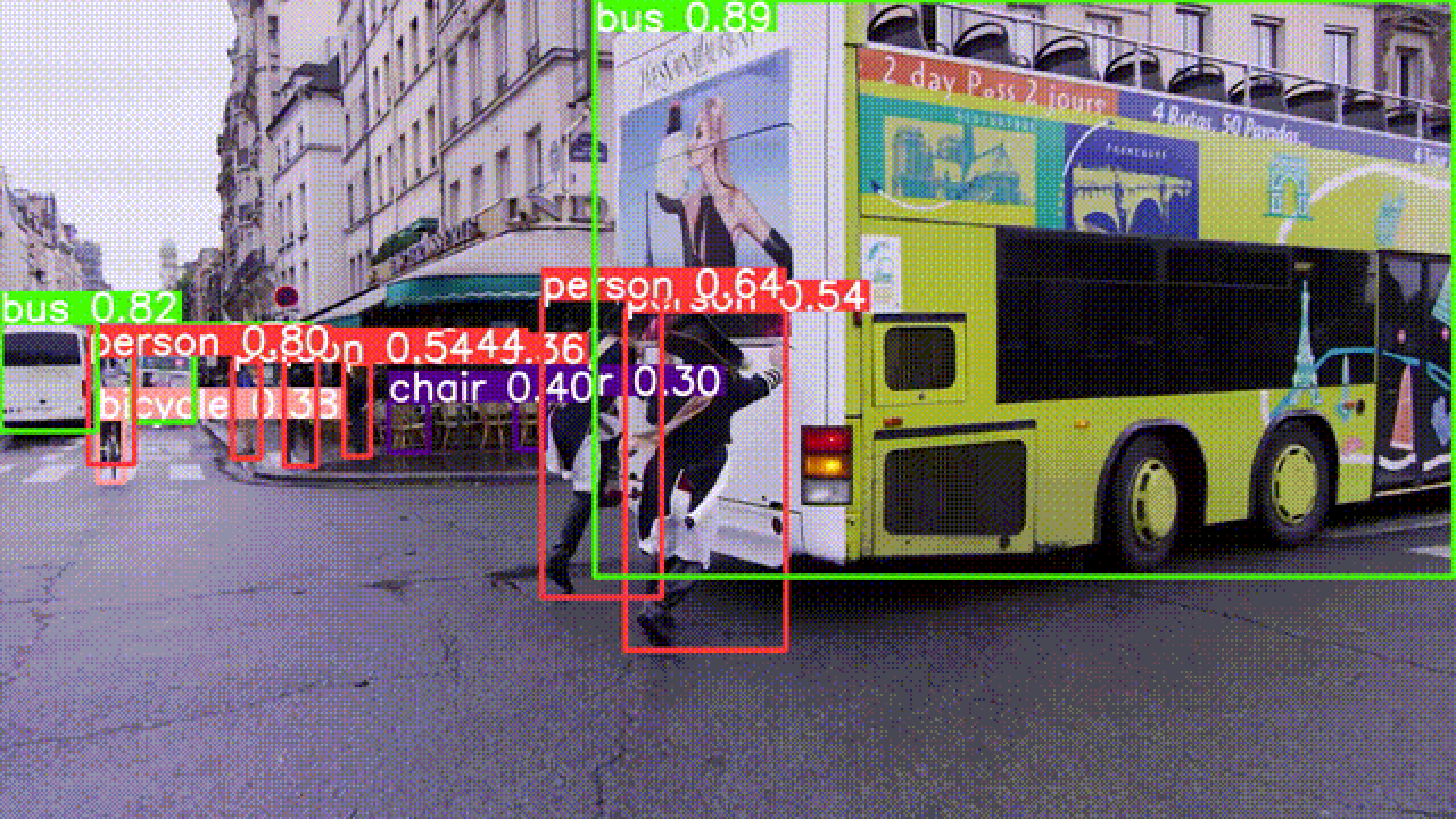


Ren, He, Girshick, and Sun, 2015

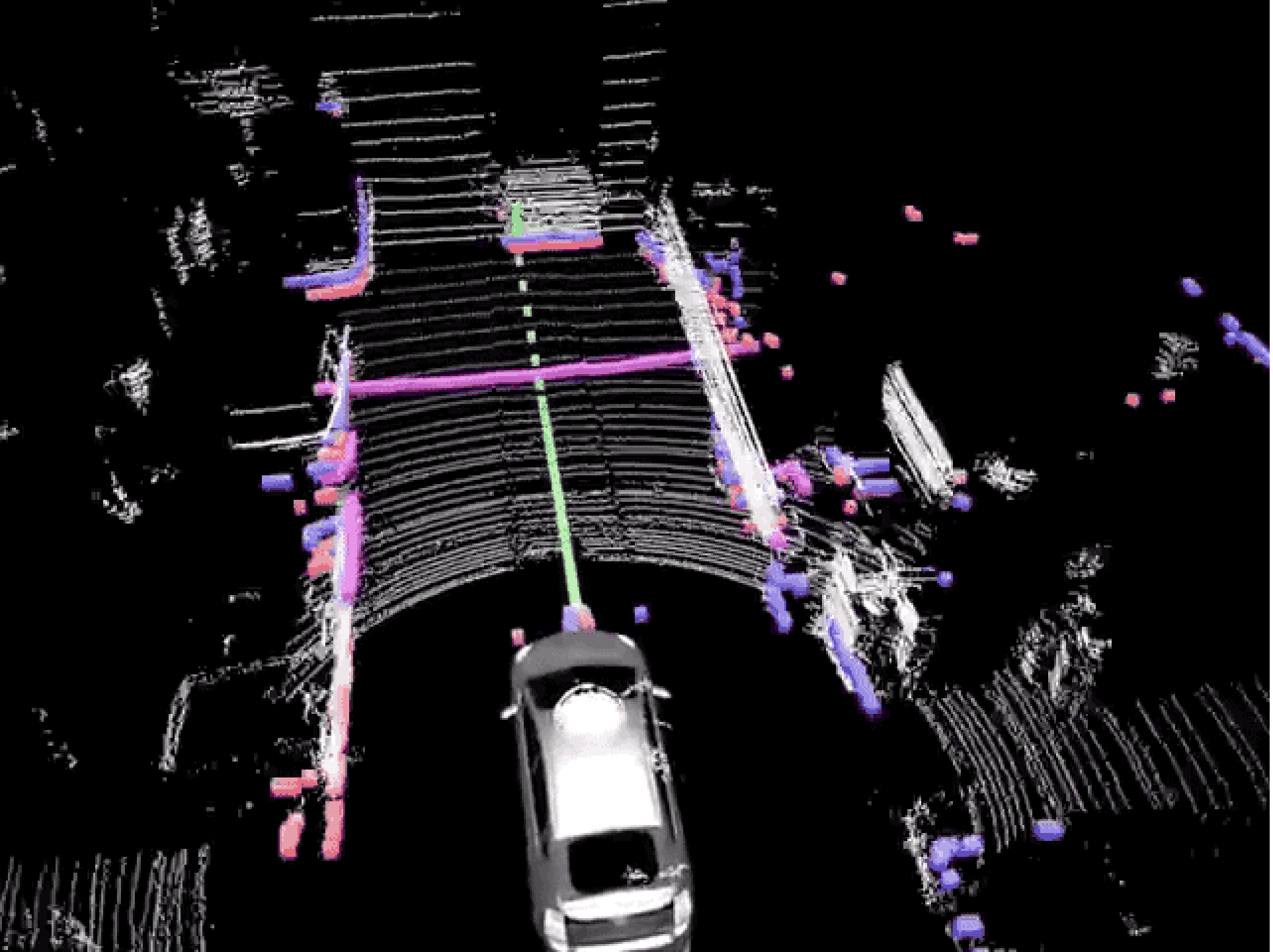
Image Segmentation

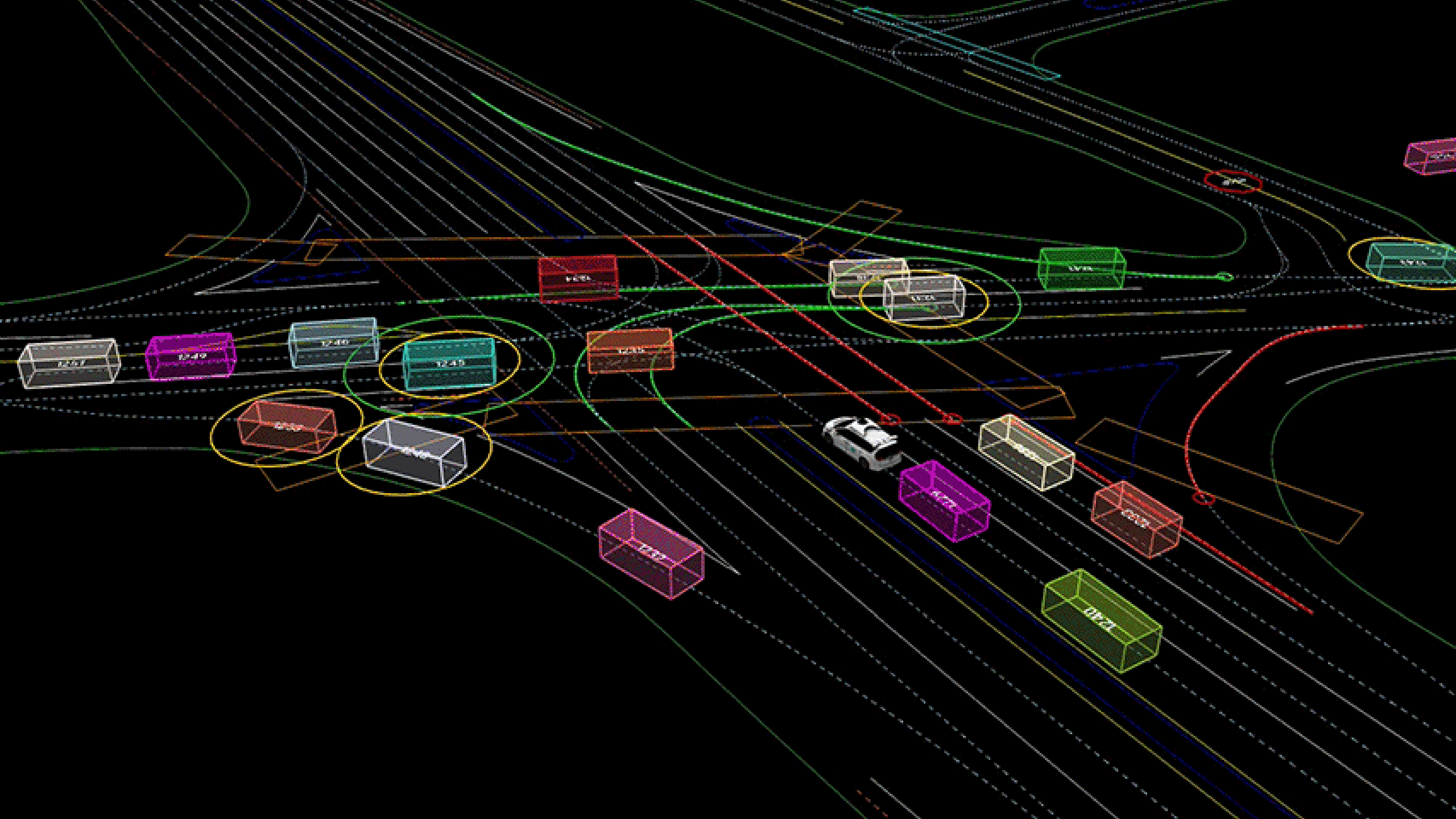


Fabaret et al, 2012


bus 0.89
2 day Pass 2 jours
bus 0.82
person 0.80
person 0.64
person 0.54
chair 0.40
0.30

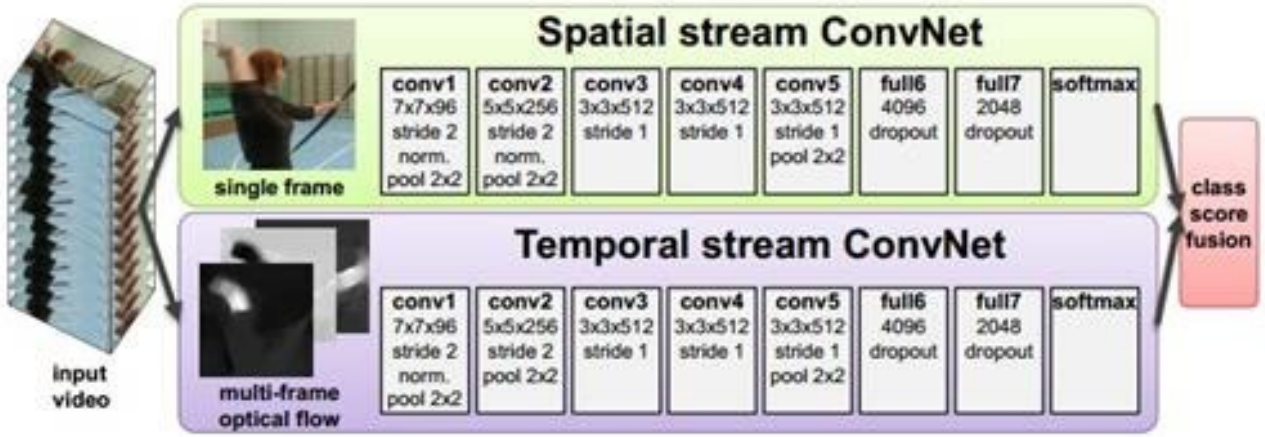

16/25 mph
AUTO





2012 to Present: Deep Learning is Everywhere

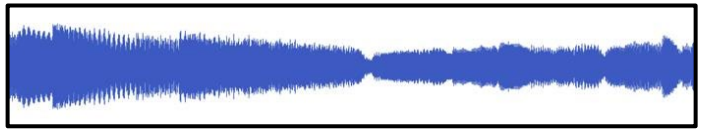
Video Classification



Simonyan et al, 2014

Multimodal Video Understanding





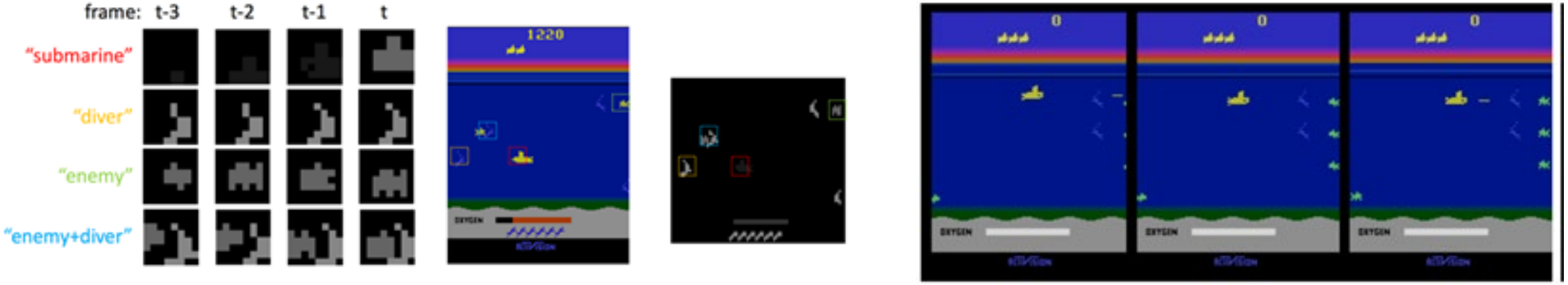
Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)



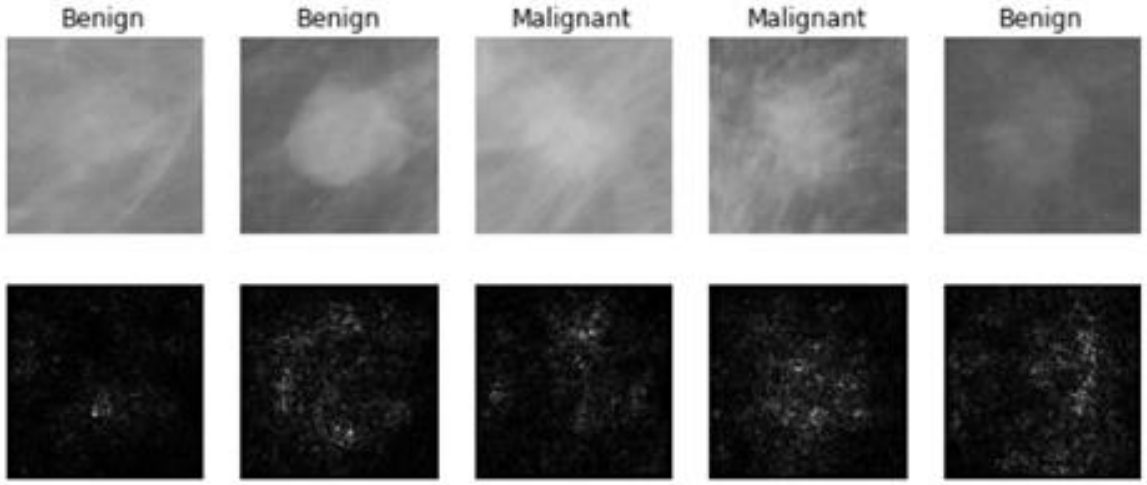
Playing Atari games (Guo et al, 2014)



Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Medical Imaging (Levy et al, 2016)



Galaxy Classification (Dieleman et al, 2014)



Whale Recognition



Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Image Captioning

(Vinyals et al, 2015

Karpathy and Fei-Fei, 2015)



A white teddy bear sitting in the grass



A man in a baseball uniform throwing a ball



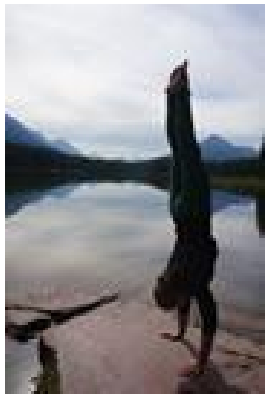
A woman is holding a cat in her hand



A man riding a wave on top of a surfboard



A cat sitting on a suitcase on the floor

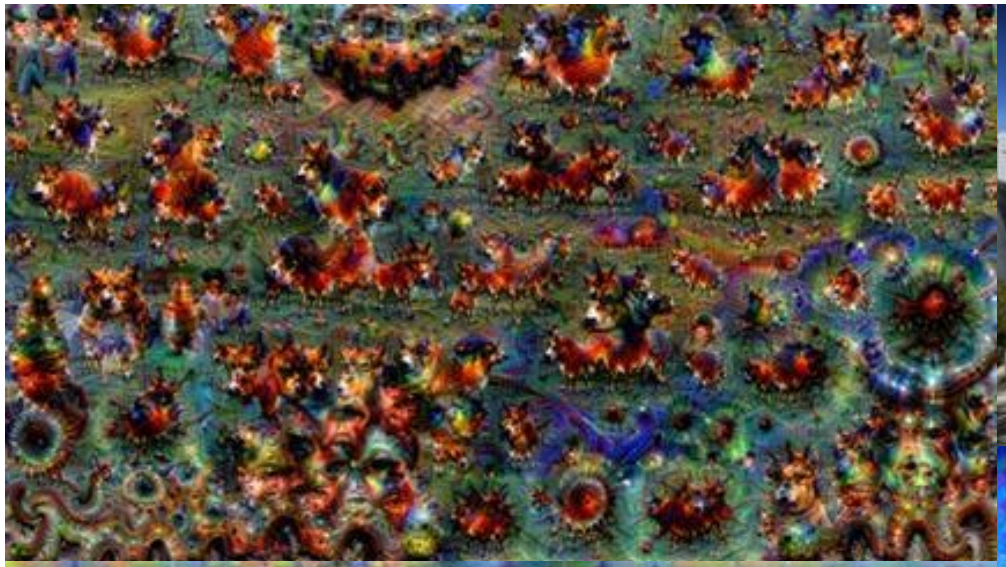


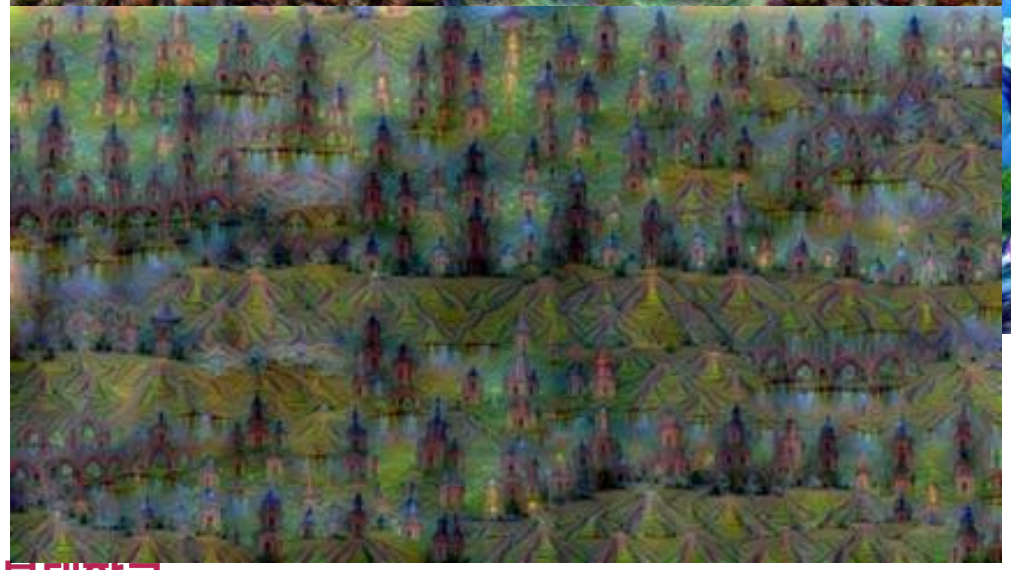
A woman standing on a beach holding a surfboard

Slide inspiration: Justin Johnson

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Beyond 2D Recognition: Generative Modeling













Style Transfer

Mordvinsev et al, 2015

Gatys et al, 2016

Slide inspiration: Justin Johnson

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Beyond 2D Recognition: Generative Modeling



Karras et al, “Progressive Growing of GANs for Improved Quality, Stability, and Variation”, ICLR 2018

Slide inspiration: Justin Johnson

Beyond 2D Recognition: Generative Modeling



"Teddy bears working on new AI research underwater with 1990s technology"

DALL-E 2

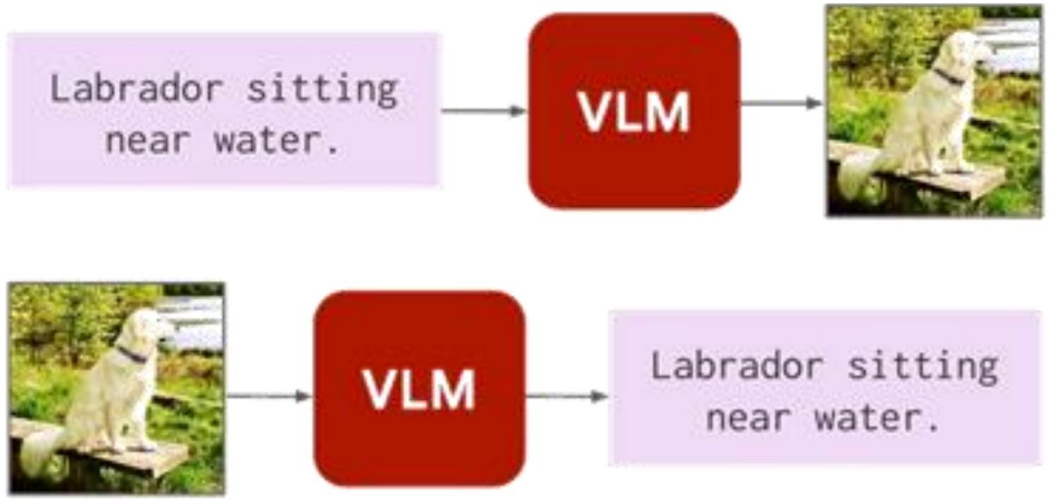


"Rendered in an intricate origami art style using complex, angular folds and crisp creases. A multilayered diorama depicts a cute neighborhood street entirely from folded paper – houses with sharp rooflines, precise white picket fences, and layered, geometric flowers and rose bushes in vibrant paper hues. Focused lighting enhances the dimensionality. A vibrant origami cat, its body segmented by distinct, sharp folds, moves with articulated, deliberate steps along the paper sidewalk. Its limbs shift segment by segment, maintaining crisp creases as it progresses. The viewpoint tracks smoothly alongside the cat, revealing successive layers of the detailed papercraft neighborhood scrolling past, enhancing the scene's geometric depth and dimensionality."

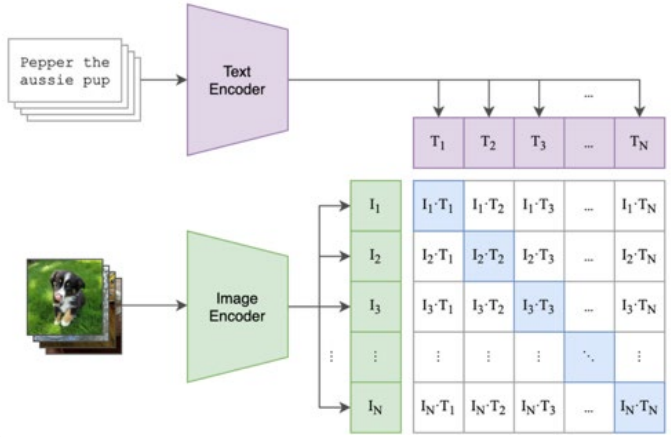
Veo

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Beyond 2D Recognition: Vision Language Models



Yasunaga, Michihiro, et al. "Retrieval-augmented multimodal language modeling." arXiv preprint arXiv:2211.12561 (2022).



Contrastive pre-training in CLIP. The blue squares are the pairs for which we want to optimize the similarity. Image derived from https://github.com/openai/CLIP



OpenCV

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Beyond 2D Recognition: 3D Vision



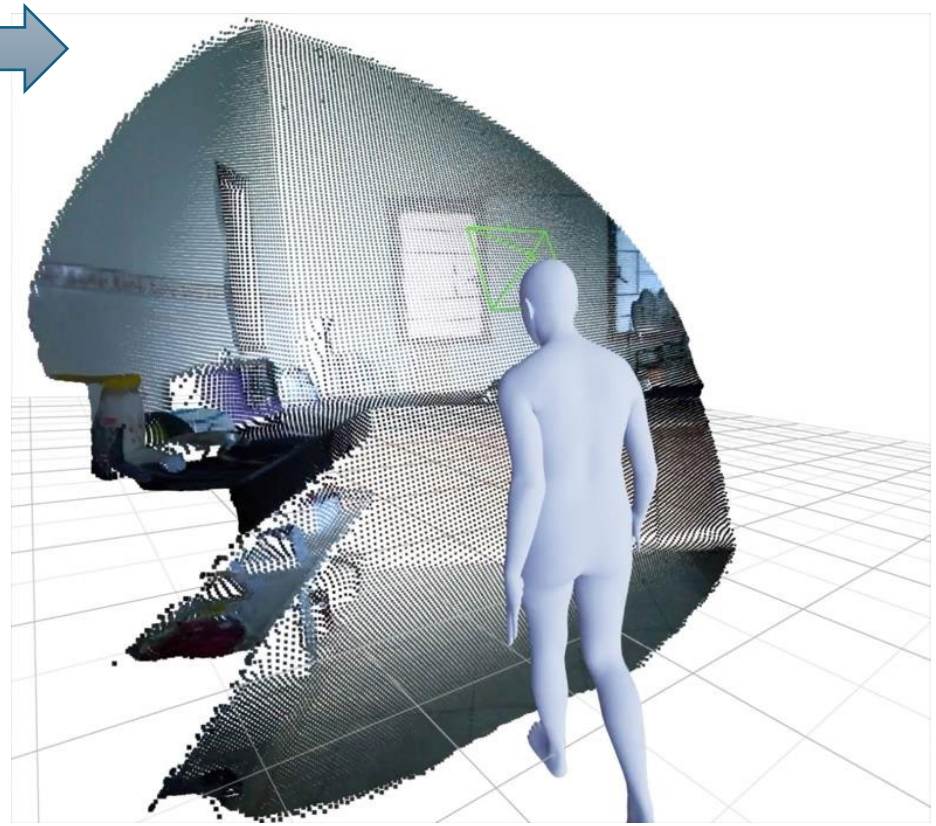




Choy et al., 3D-R2N2: Recurrent Reconstruction Neural Network (2016)



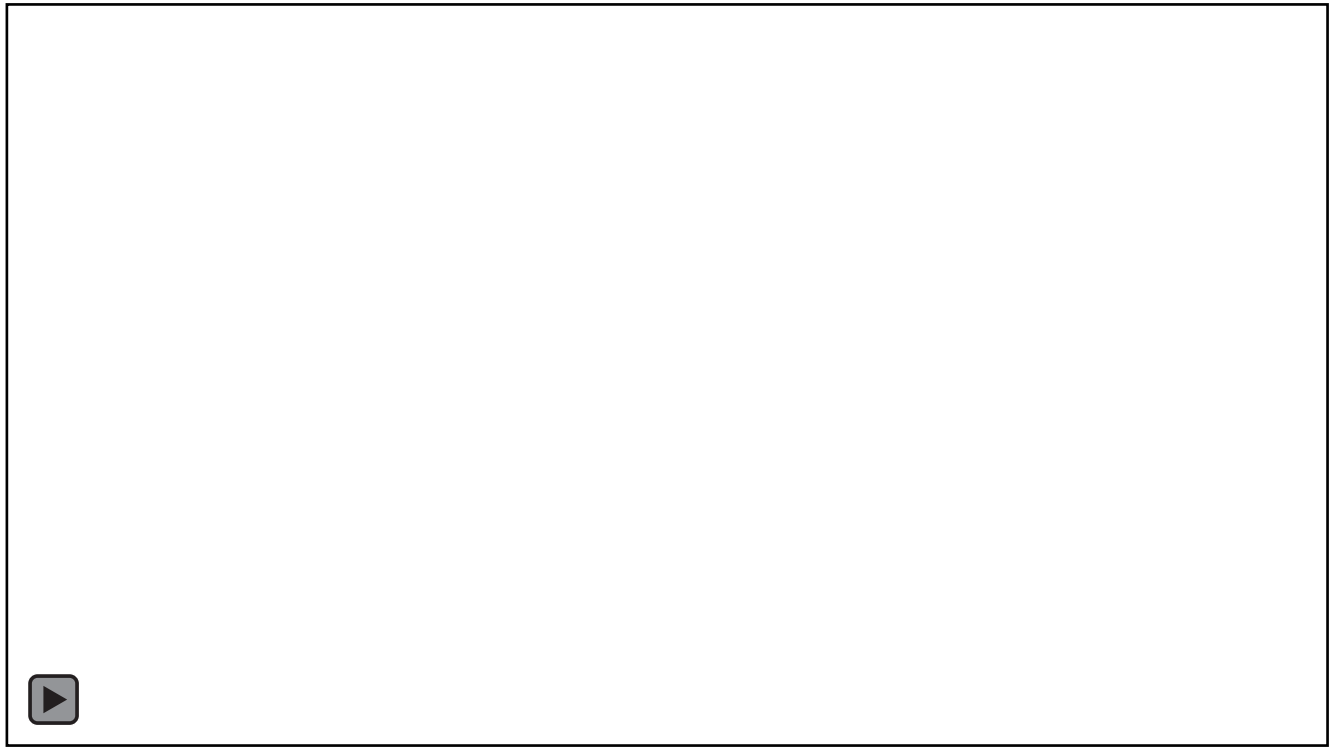
Zhou et al., 3D Shape Generation and Completion through Point-Voxel Diffusion (2021)



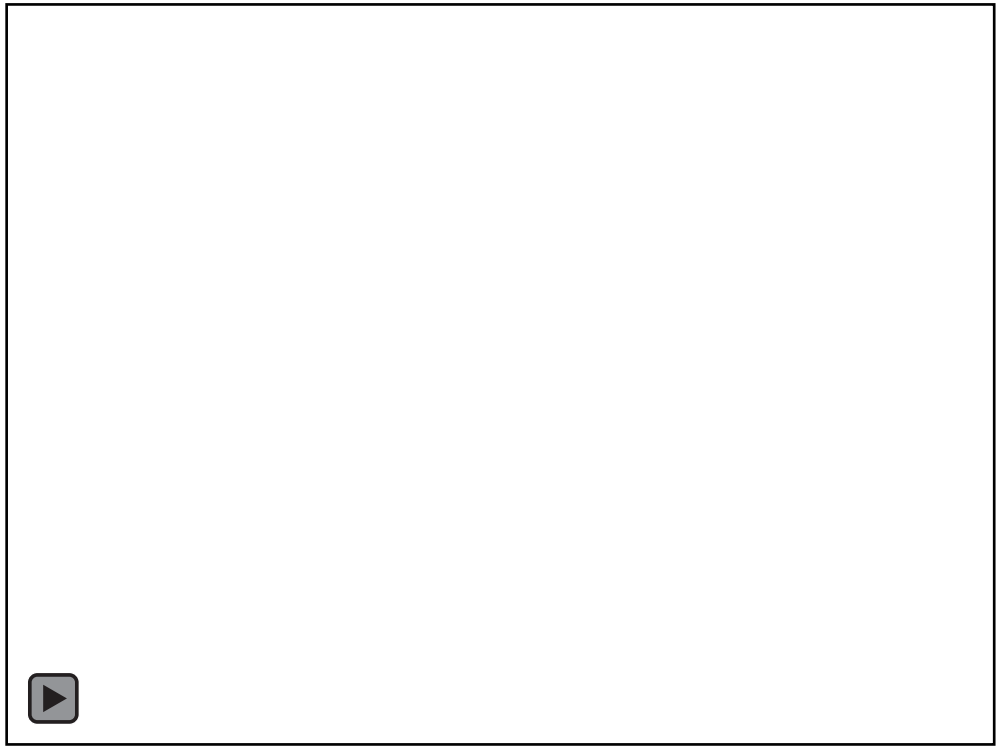
Patel et al. (2026)

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Beyond 2D Recognition: Embodied Intelligence



https://www.pi.website/blog/pi05

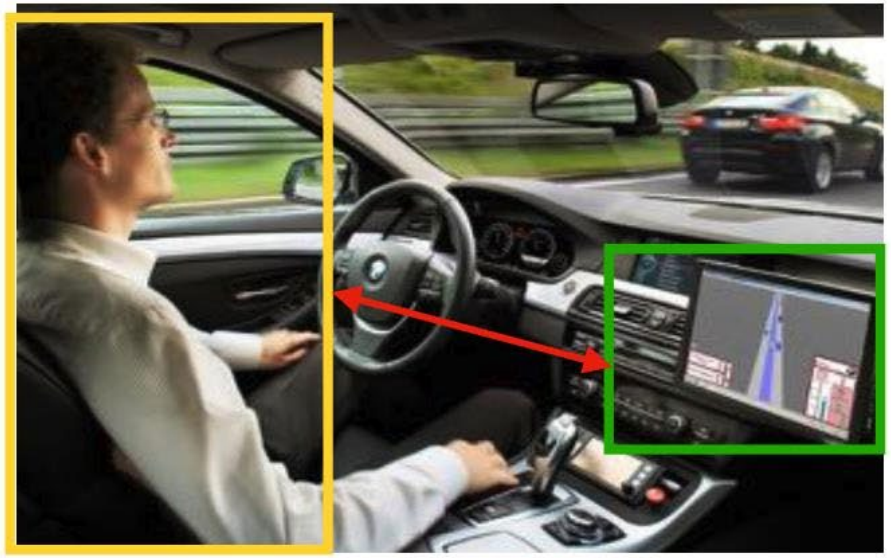


Mandlekar and Xu et al., Learning to Generalize Across LongHorizon Tasks from Human Demonstrations (2020)

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

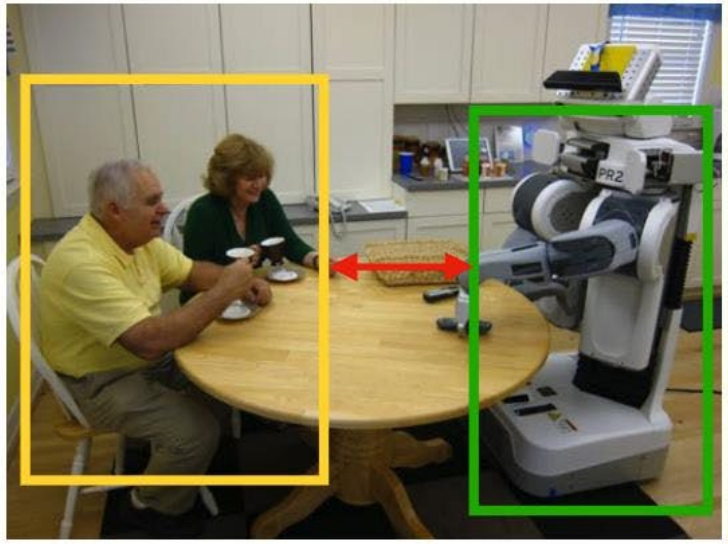
Human-Centered Applications and Implications

Self-Driving



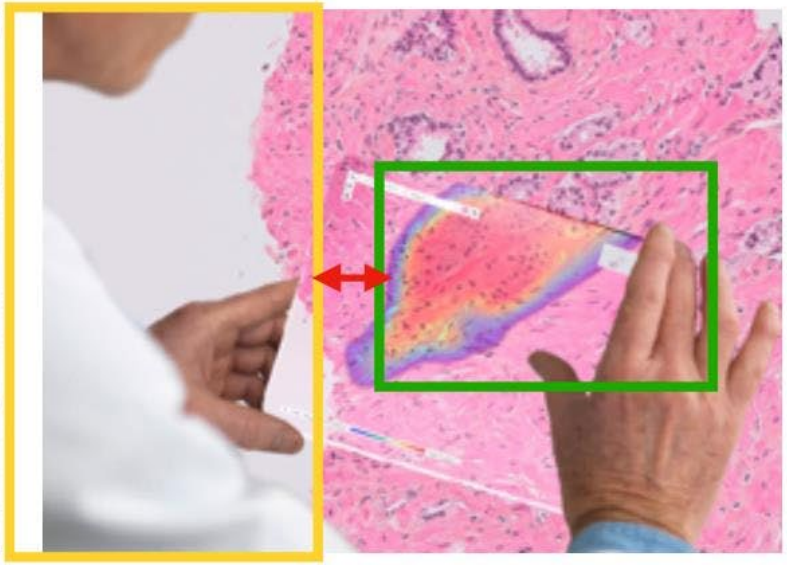
Driver and Autopilot

Home Robot



Elders and Robot

Healthcare

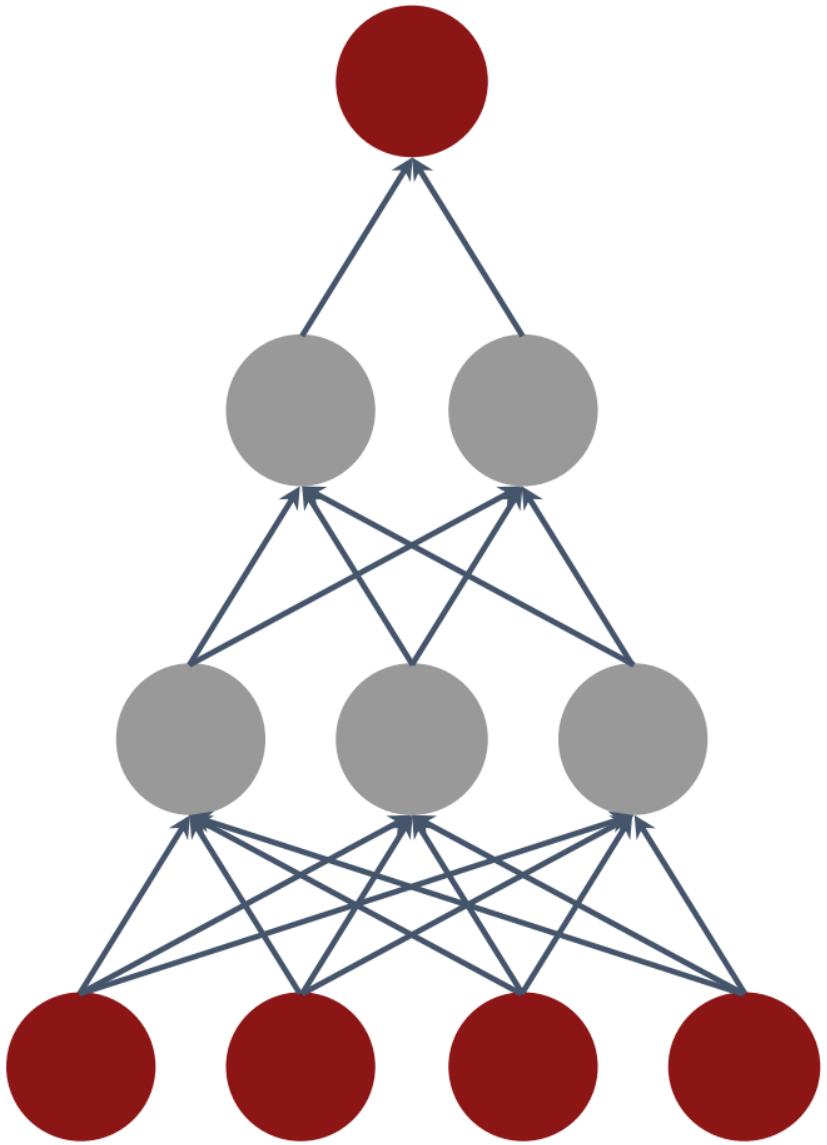


Physician and Medical AI

build trustworthy interactions between humans and AI

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

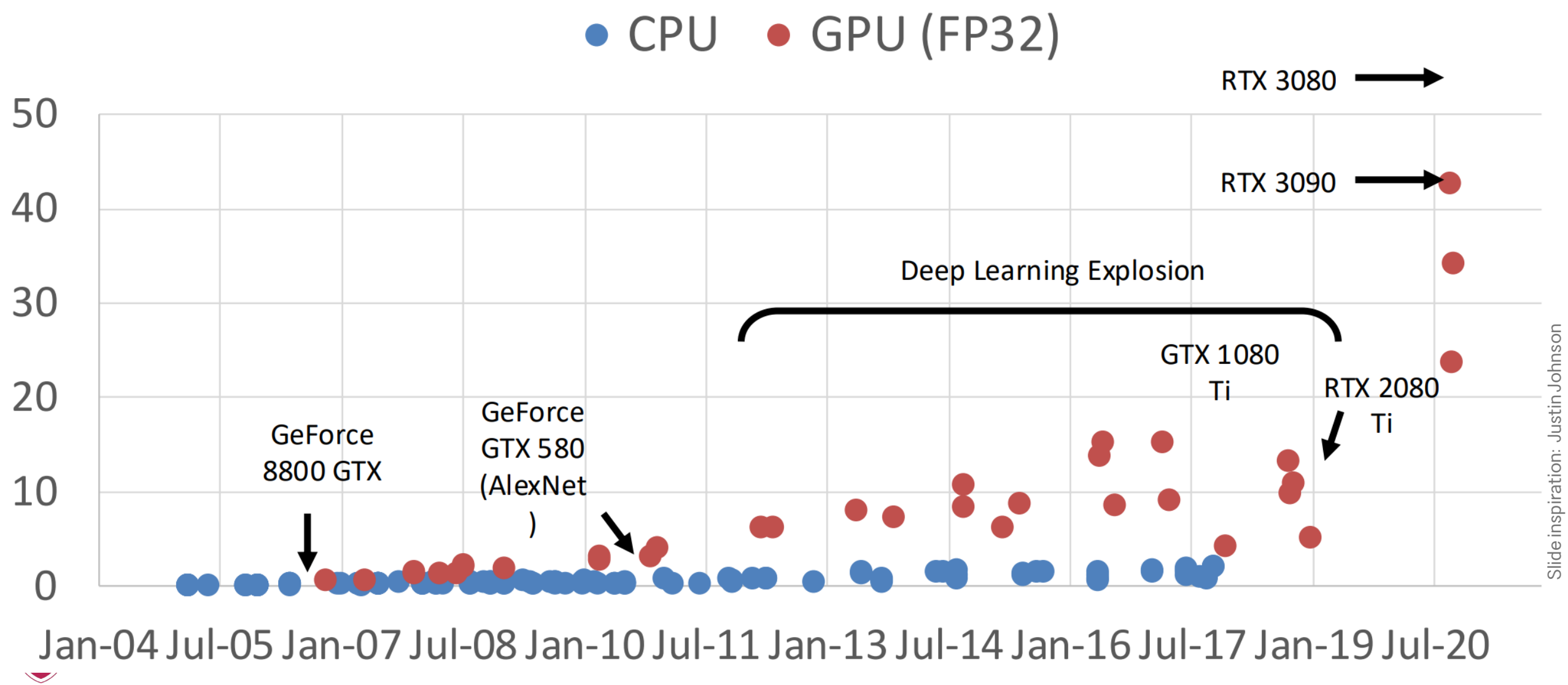




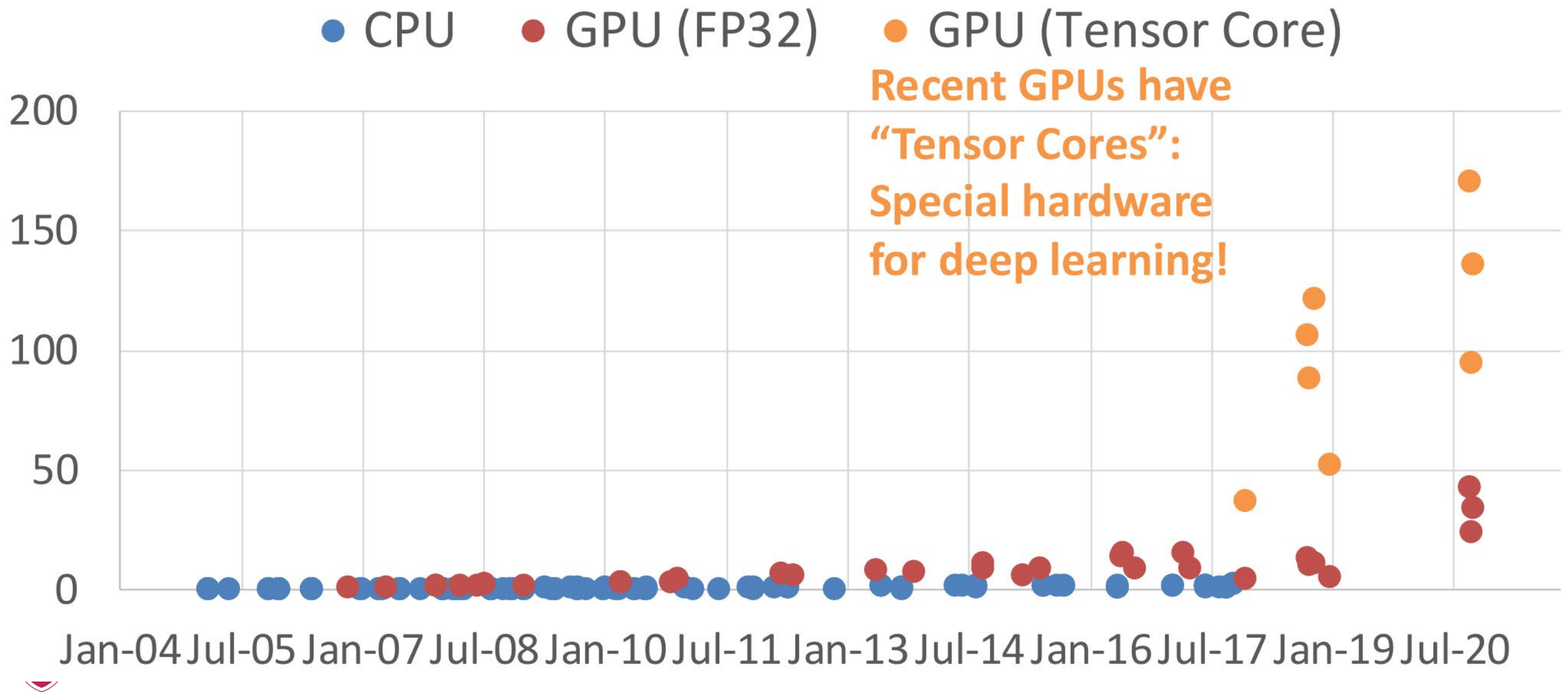
Algorithms



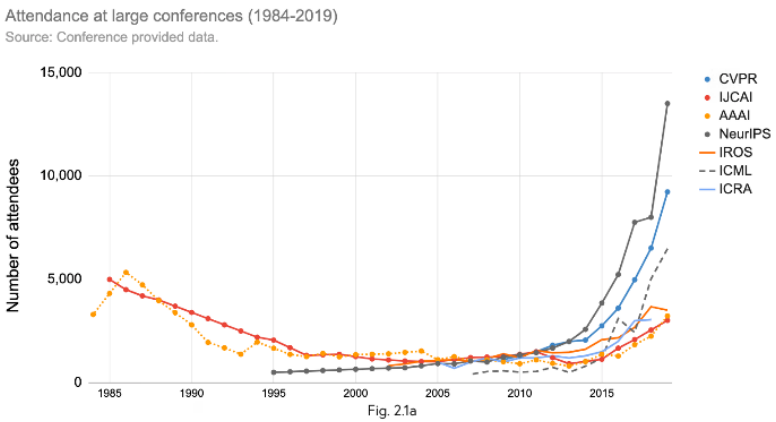
GFLOP per Dollar



GFLOP per Dollar

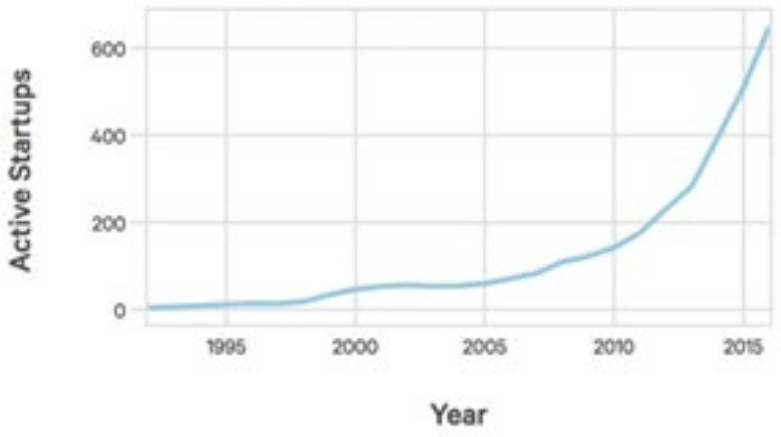


AI's Explosive Growth & Impact



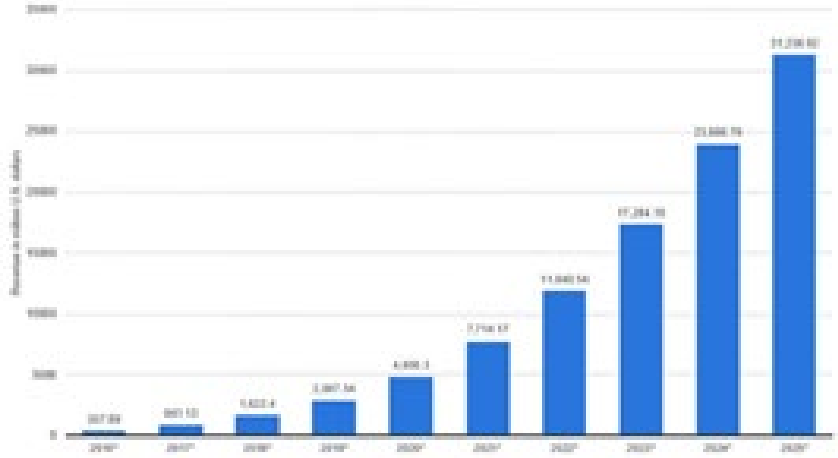
Number of attendance at AI conferences

Source: The Gradient



Startups Developing AI Systems

Source: Crunchbase, VentureSource, Sand Hill Econometrics



Enterprise Application AI Revenue

Source: Statista

Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Despite the successes, computer vision still has a long way to go

Computer Vision Can Cause Harm

Harmful Stereotypes

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

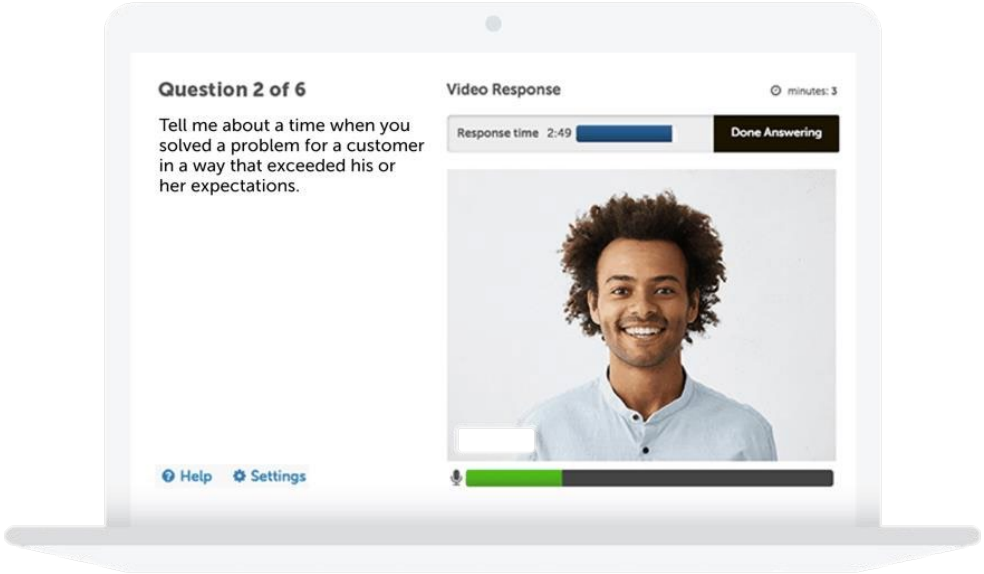


Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante

Computer Vision Can Save Lives



Slide inspiration: Fei-Fei Li & Ehsan Adeli & Zane Durante